Team 96 — Formula 1 Podium Prediction — Milestone 1 Report

We predict whether a driver finishes on the podium using only information available after qualifying and before the race starts. On the unseen 2022–2024 test seasons, all three models reach ROC-AUC ≈ 0.94 and PR-AUC 0.69–0.72, with Gradient Boosting strongest on test (PR-AUC 0.724, F1 0.697).

1. Introduction

The task is binary classification: does a driver finish on the podium (`positionOrder` ≤ 3)? Predicting the exact finishing position is out of scope.

Prediction cutoff. Every predictor must be known after qualifying and before the race starts. Anything produced during or after the race (finishing position, status, laps, points, fastest lap, pit stops) is excluded from the features.

Data. The Kaggle / Ergast *Formula 1 World Championship (1950–2024)* dataset: 14 relational CSV tables covering 75 seasons, 1,125 races and 26,759 raw result rows. Answering any question requires joining several tables.

Grain. The modeling table has exactly one row per driver per race, i.e. a unique `(raceId, driverId)` pair.

Approach. We audit and clean all tables, answer three multi-join data-engineering questions, engineer four groups of leakage-safe features, and compare Logistic Regression, Gradient Boosting and a shallow neural network on a chronological train / validation / test split. We then run feature-group ablation and explainability analyses and package the selected model as an inference function.

2. Data audit and exploratory analysis

The raw audit found structurally sound keys but three issues that drive the cleaning: text missing-value sentinels, duplicate driver-race rows, and strong changes in data coverage over time. All raw tables were left untouched; every later step works on copies.

Audit check	Finding
Missing values	Missingness is stored as the text sentinel `\N` (about 160,000 cells; results alone 122,887), so affected columns load as strings
Primary keys	All 14 tables unique and non-null (composite keys for lap times and pit stops)
Foreign keys	Every foreign-key link checked, 0 orphan rows
Driver-race grain	85 duplicate `(raceId, driverId)` pairs over 176 rows, almost all 1950–1964 shared drives
`grid = 0`	1,638 rows, mostly non-starters (did not qualify / prequalify, withdrew)
Target	Raw podium rate 12.69% — an imbalanced binary target

EDA findings

The sport changed so much across 75 seasons that raw history cannot be modelled as one homogeneous period.

- **Races per season** grew from 7 in 1950 to 24 in 2024.

- **Field size** peaked around 39 entrants per race in 1989 because of pre-qualifying and non-qualifiers; today about 20 cars start.
- **Qualifying coverage** is 0% before 1994, patchy at 6–59% in 1996–2002, and complete from 2003. This drift determines the training window (Section 6).
- **Grid position** has a steep, non-linear relationship with podium probability, motivating grid as a core feature.

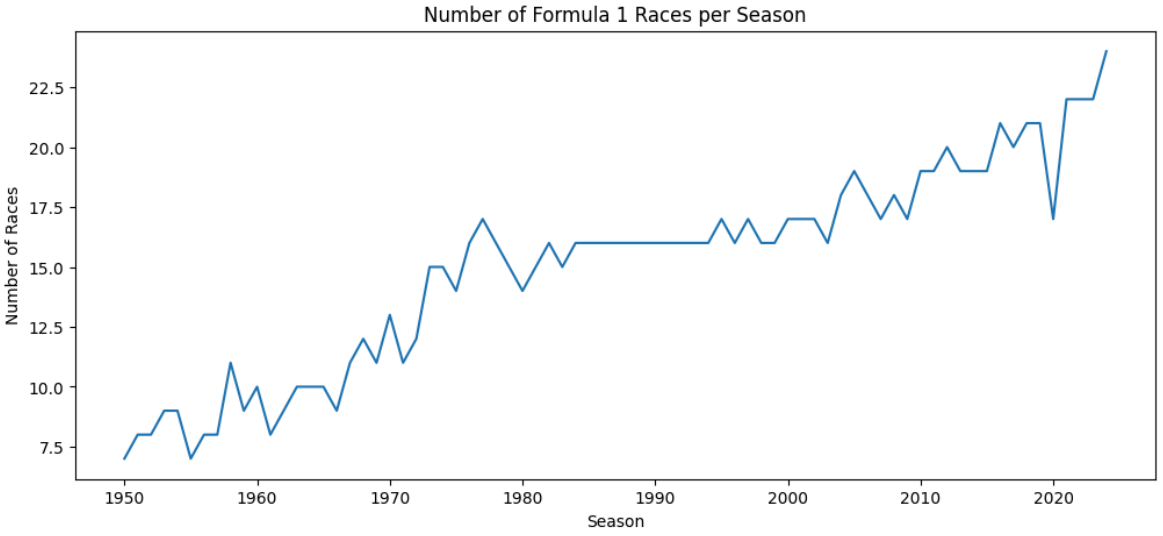


Figure 1. Number of races per season

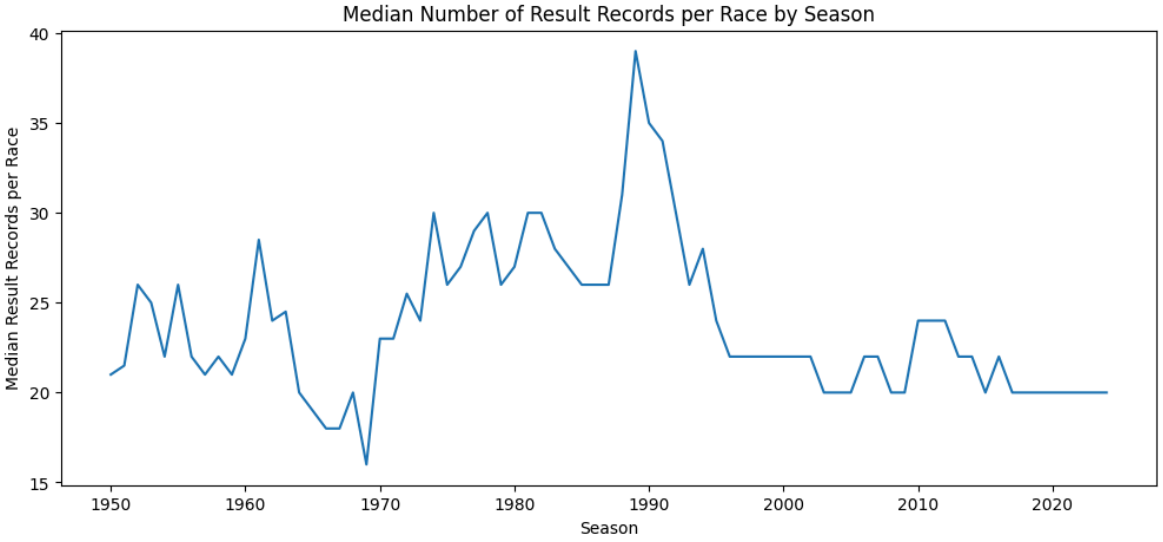


Figure 2. Median result records per race by season

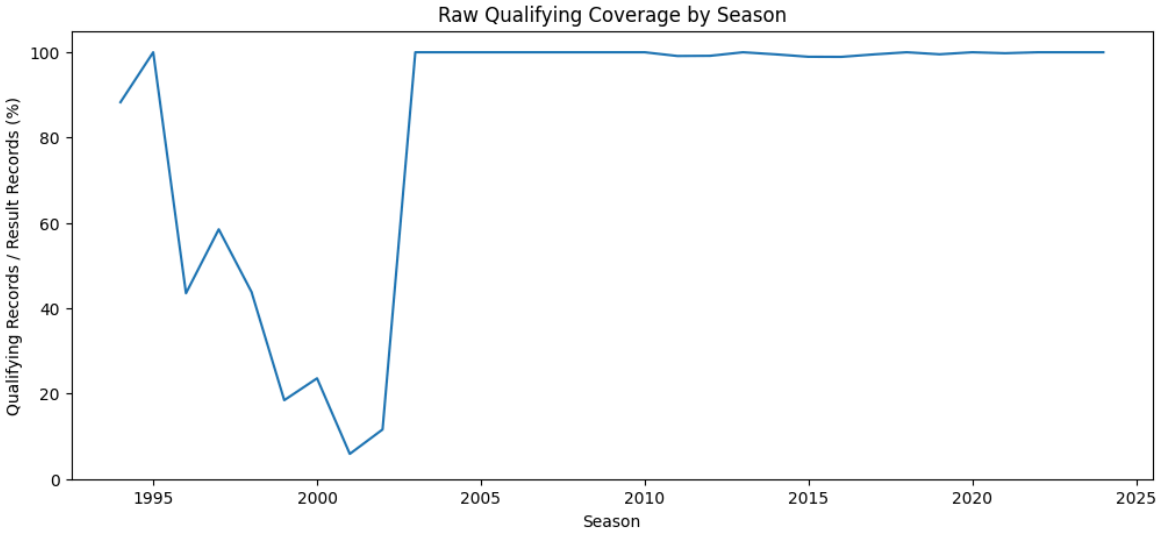


Figure 3. Raw qualifying coverage by season

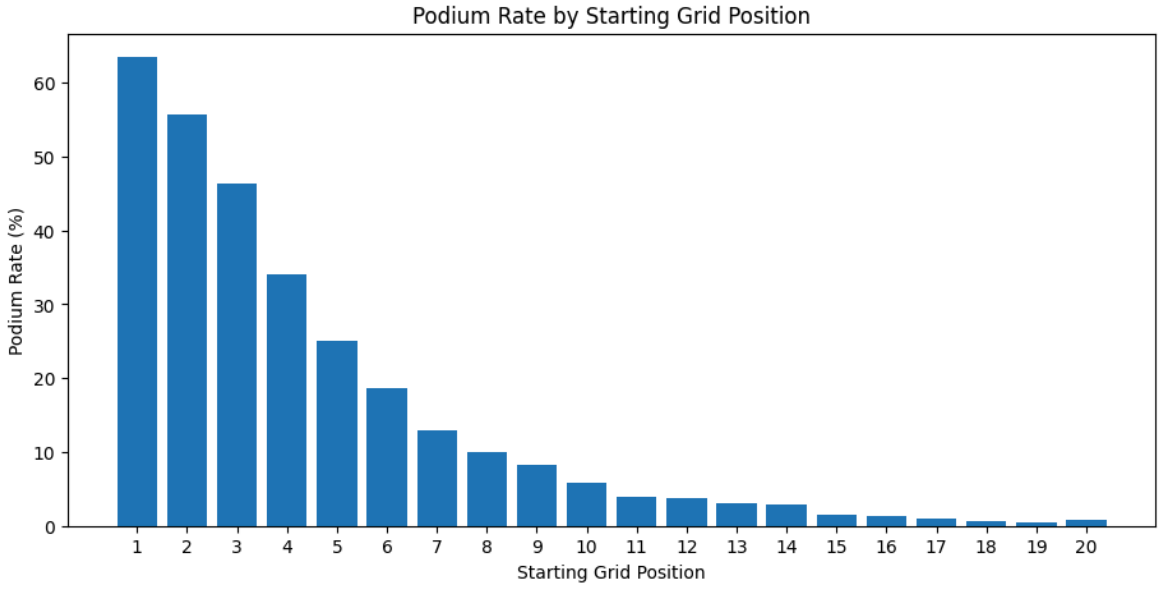


Figure 4. Podium rate by starting grid position

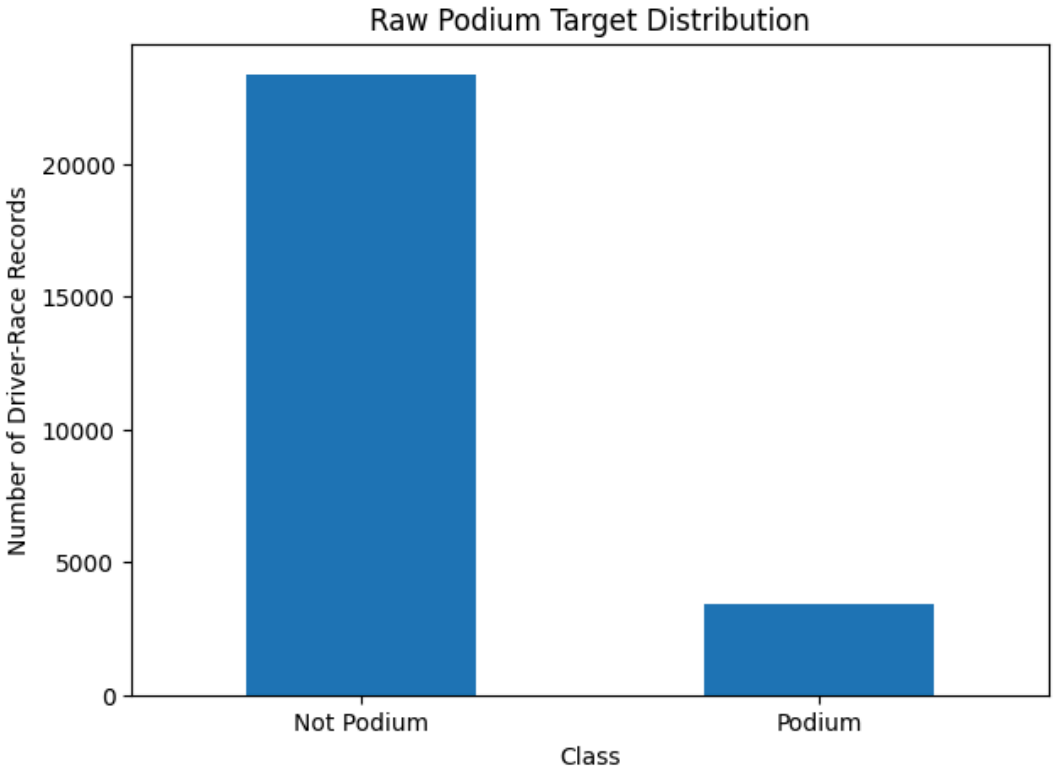


Figure 5. Raw podium target distribution

3. Data cleaning

Cleaning reduced the modeling cohort from 26,759 raw result rows to 24,643 driver-race rows (−7.9%) and resolved all 85 duplicate pairs, while the podium rate moved only from 12.69% to 13.63%.

Cleaning happens in **two layers**. Layer 1 cleans all 14 tables without dropping rows; the data-engineering questions use it. Layer 2 builds the modeling table from Layer 1. Keeping them separate means scope decisions for the model never silently change the historical answers.

Step	What we did	Why
Sentinels	Converted every `\N` to a true missing value; validated that none remain	As text, numeric columns cannot be computed on and missing values are not counted as missing
Semantic typing	Dates to datetime; lap and pit times like `1:23.456` to seconds; races ordered by `(year, round)`	Times must be numeric to compare; every "past races only" feature depends on correct chronological order
Structural missingness	Kept empty Q2/Q3 as missing and added `q2_recorded` / `q3_recorded` flags	An empty Q3 means the driver was eliminated, which is information, not noise
Keys	Re-validated primary and foreign keys after cleaning	Broken links would silently drop or duplicate rows in joins
Non-starters	Removed did not qualify / prequalify / withdrew rows and F/W rows without a grid slot (1,647 rows); kept 87 real pit-lane starts flagged as `grid_zero_start`	At the cutoff we already know these drivers will not race; they are trivial negatives that inflate scores
Failed-to-start	Kept W-coded drivers who had a grid slot	At the cutoff their failure on the formation lap is not yet known; removing them would leak future information

Step	What we did	Why
Indianapolis 500	Excluded the 11 races (405 rows, 1950–1960) from the modeling cohort	A largely separate race with a different field; it distorts grid and home-race patterns
Shared drives	Kept the row with the best `positionOrder` per `(raceId, driverId)`	Grid cannot identify the original car (both rows have a grid value in almost all pairs). Keeping the first row preserves only 6 of 21 podiums; best finish preserves all 21

Validation asserts confirm one row per `(raceId, driverId)`, no missing target, and at most three podium *positions* per race. Fifteen 1950–1960 races legitimately have more than three podium drivers because co-drivers of one car are officially credited with its finish.

The shared-drive rule uses the race outcome, so it is a historical deduplication rule, not a feature. All affected rows are from 1950–1964, outside the 2003–2024 modeling period, so it cannot leak into training or evaluation.

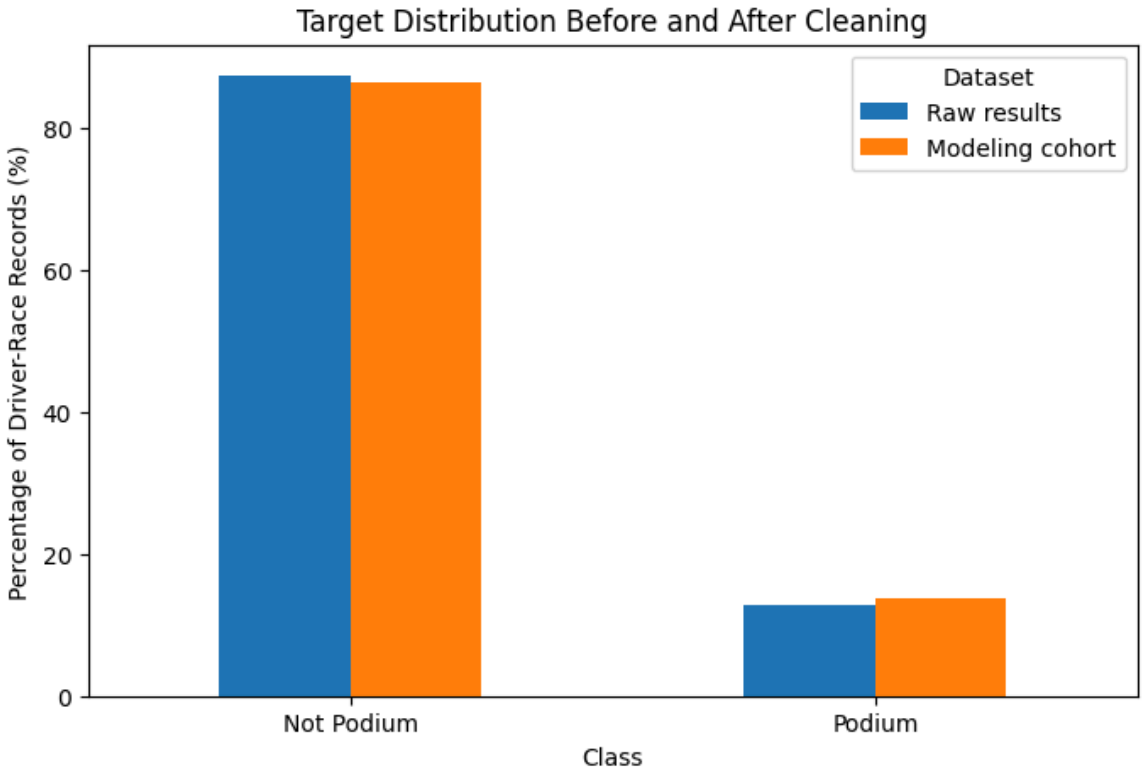


Figure 6. Target distribution before and after cleaning

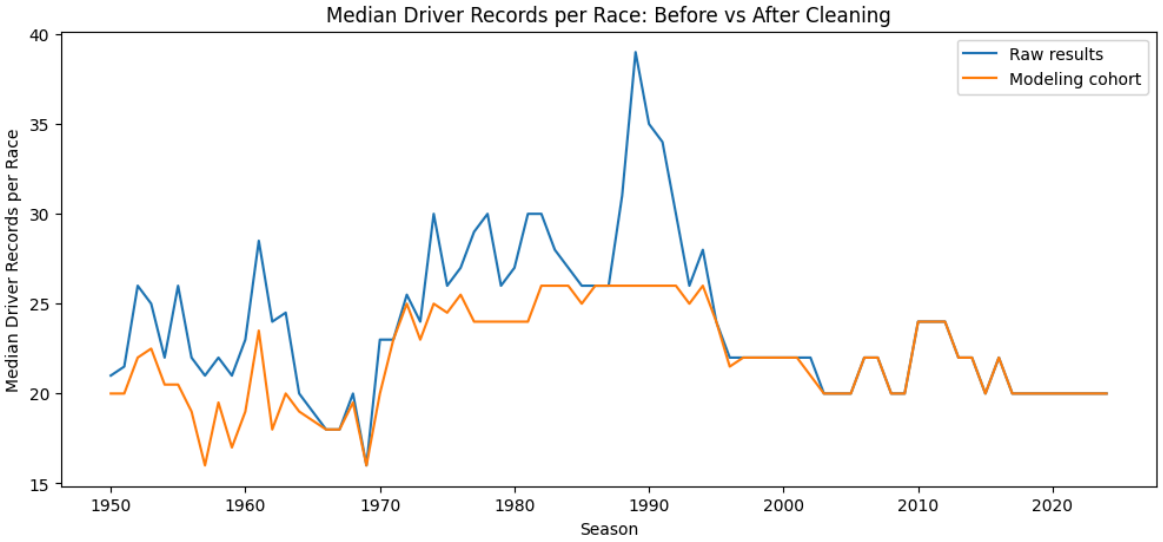


Figure 7. Median driver records per race before vs after cleaning

4. Data-engineering questions

All three questions use the cleaned driver-race cohort and multi-table joins; one row is always one driver in one race.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Answer. The ranking changed considerably after the 2014 hybrid era. Among the 14 circuits with at least 10 front-row starts in both 2003–2013 and 2014–2024, Monaco and Monza led the earlier period (77.3%), while Suzuka (88.9%), Yas Marina (86.4%) and Monaco (85.0%) led 2014–2024. The Spearman rank correlation between the two eras is only about 0.30 ($p = 0.29$).

Qualifying vs grid. Qualifying and grid position are not interchangeable: excluding the 2021–2022 sprint weekends, they matched exactly only 73.15% of the time. 32 drivers qualified on the front row but started further back, and 31 were promoted onto it. With a qualifying-defined front row the era correlation rises to about 0.55 ($p = 0.043$), so penalties between qualifying and the grid explain part of the ranking change. The actual grid is the primary definition because the question concerns where drivers started.

Tables joined: results, races, circuits, qualifying, sprint_results.

- *Grain:* one driver in one race.
- *Missing data:* qualifying is used only from 2003, where coverage is complete; the six 2021–2022 sprint weekends are excluded from the qualifying version because qualifying set the sprint grid, not the race grid.
- *Pre-race cutoff:* qualifying position and grid are pre-race; finishing position is used only as the historical outcome.

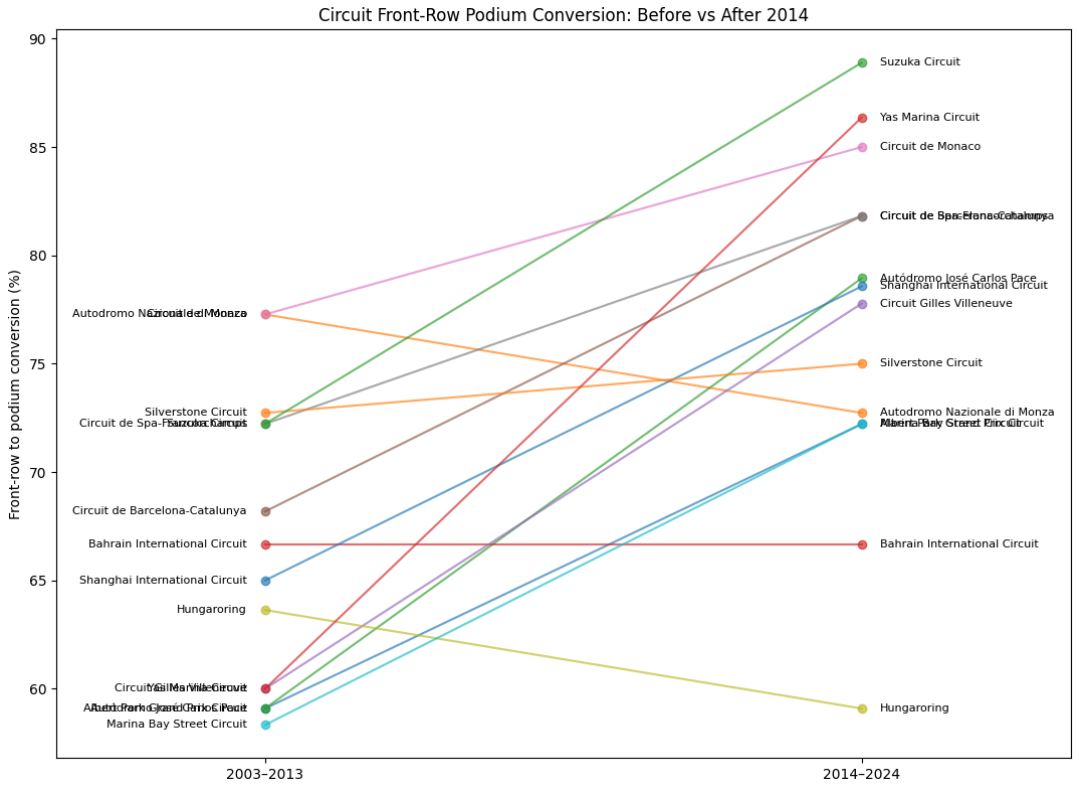


Figure 8. Circuit front-row podium conversion before vs after 2014

Q2. Do drivers podium more often at home, after controlling for starting position?

Answer. No. Home drivers had slightly higher podium rates from grid groups 1–3 and 4–6 but slightly lower rates from 7–10 and 11+. After adjusting for grid group with logistic regression, the home-race odds ratio is 0.955 (95% CI 0.806–1.132, p = 0.597). The interval includes 1, so there is no evidence of a home advantage once starting position is accounted for.

Tables joined: results, drivers, races, circuits.

- *Grain:* one driver in one race.
- *Missing data:* nationalities were mapped to circuit countries (`USA` and `United States` merged, trailing spaces stripped); none remained unmapped. Hyphenated nationalities count as home in either country. The 87 grid-zero starts were excluded because they fit no grid group.
- *Pre-race cutoff:* grid and nationality are pre-race; podium is the outcome only. This is an association, not a causal effect.

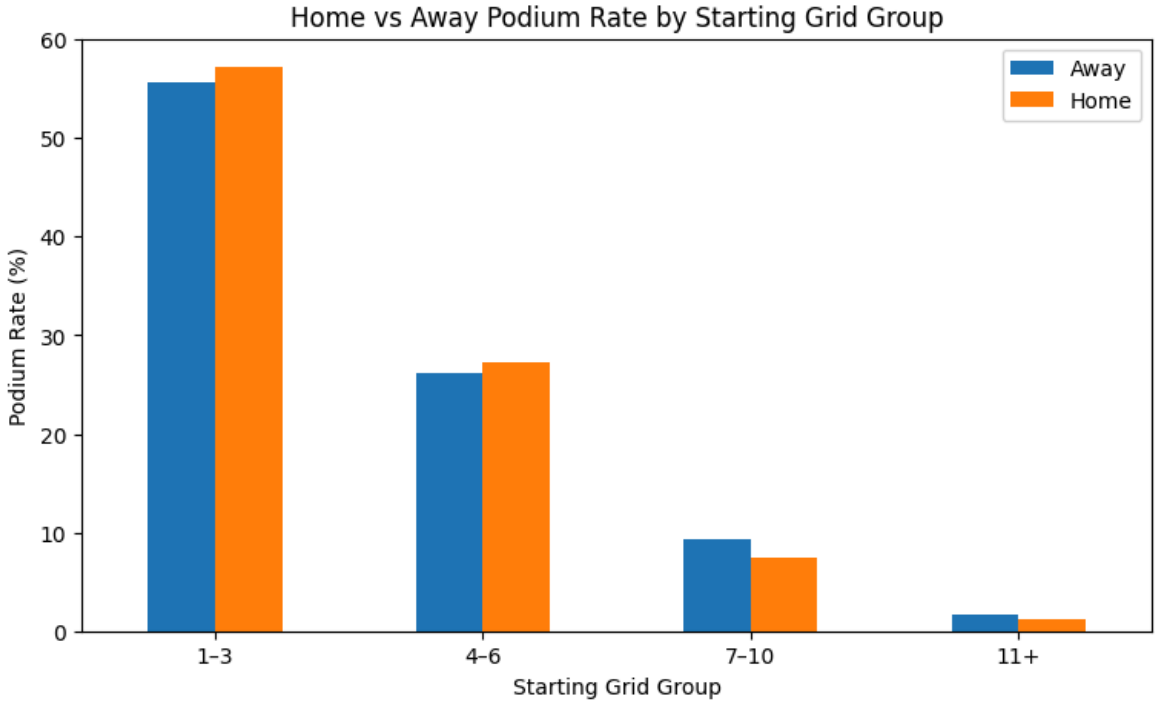


Figure 9. Home vs away podium rate by starting grid group

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Answer. Reliability improved: the overall strict mechanical-retirement rate fell from 8.85% of car starts in 2014–2021 to 5.40% in 2022–2024. Among constructors with at least 60 starts, Lotus F1 (17.3%), Toro Rosso (16.1%) and Renault (13.1%) were highest in 2014–2021; Alfa Romeo (12.5%), Alpine (10.4%) and AlphaTauri (6.8%) were highest in 2022–2024. Caterham had the highest raw 2014–2021 rate (21.2%) but from only 33 starts.

Classification. Statuses were split into strict mechanical (engine, gearbox, hydraulics, power unit, etc.), clearly non-mechanical (collisions, accidents, finished), and ambiguous (Retired, puncture, wheel, wing damage). An assert confirms no status is left unclassified. Rates use car starts, not counts, because the eras differ in length; drivers who withdrew before the start (14) are excluded.

Robustness. Adding the ambiguous statuses barely changes the 2014–2021 ranking (rank correlation 0.96) but changes the 2022–2024 ranking more (0.63), so lower 2022–2024 positions should be read cautiously.

Tables joined: results, status, races, constructors.

- *Grain:* one car start in one race.
- *Missing data:* ambiguous statuses are handled by the sensitivity analysis; renamed teams follow `constructorId` and appear as separate constructors.
- *Pre-race cutoff:* status is post-race, valid here because this is a retrospective question, but never used as a model feature.

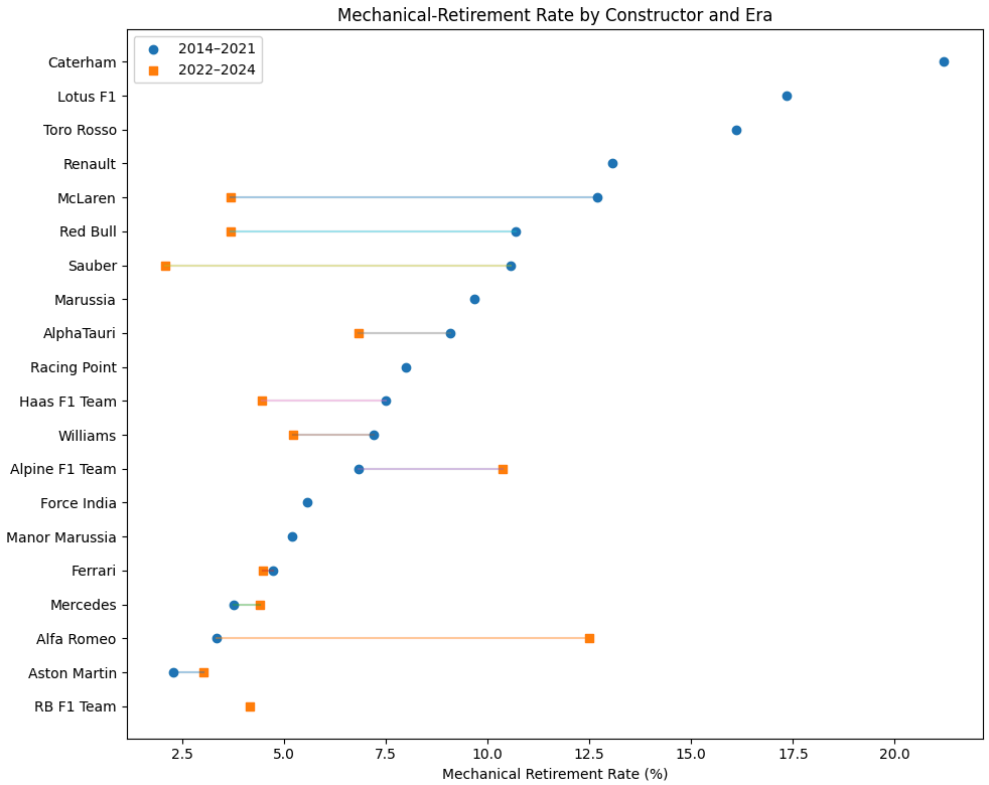


Figure 10. Mechanical-retirement rate by constructor and era

5. Selected features and their validity at the pre-race cutoff

We use 21 features in four groups — grid, qualifying, standings and form — every one of which is known after qualifying and before lights out. They were computed on the full 1950–2024 history (so early-2003 form uses 2002 races) and only then restricted to 2003–2024.

Group	Feature	Why it is valid before the race
Grid	`grid`	Starting slot is published before the race
Grid	`grid_zero_start`	Pit-lane start is known when the grid is set
Qualifying	`qualifying_position`	Qualifying ends before the race
Qualifying	`q1_gap_pct`	Gap to the fastest Q1 lap of the same race, from completed qualifying; capped at the training 99th percentile (7.6%)
Qualifying	`q2_recorded`, `q3_recorded`	Whether the driver reached Q2/Q3, known after qualifying
Qualifying	`grid_penalty`	Grid minus qualifying position; penalties are applied before the race
Qualifying	`qualifying_missing`, `qual_time_missing`	Indicators of absent qualifying data
Standings	`driver_position_before`, `constructor_position_before`	Championship position **after the previous race**, never after the current one
Standings	`driver_points_share`, `constructor_points_share`	Points before the race divided by the leader's points before the race
Standings	`driver_standing_missing`, `constructor_standing_missing`	No standings yet in round 1 of a season
Form	`driver_podium_last5`, `driver_avg_finish_last5`	Average over the driver's previous 5 starts, strictly before this race

Group	Feature	Why it is valid before the race
Form	`constructor_podium_last5`	Constructor podium rate over its previous 5 races
Form	`career_starts`	Number of earlier starts
Form	`driver_form_missing`, `constructor_form_missing`	No history yet (debut races)

Design choices.

- **Points share instead of raw points:** the scoring system changed in 2010 (a win went from 10 to 25 points) and totals grow through a season, so raw points are not comparable across years or rounds.
- **Q1 for pace:** Q1 is the only session every driver runs, so it compares like with like.
- **Excluded:** driver/constructor IDs (would memorize names rather than learn patterns), sprint results (no sprints in the training years), and every current-race outcome. An assert blocks forbidden columns from the feature list.

Leakage checks. Standings are joined on the previous race, and an assert confirms no row uses its own race. Rolling form was recomputed by hand for 300 random non-zero driver-form rows and 300 constructor-form rows; all matched the stored values.

Evidence for the chosen features

On the 2003–2019 training set every group shows a strong, monotonic relationship with podium probability.

Feature (bucket)	Podium rate
Grid P1 / P2 / P3	80.3% / 62.7% / 51.6%
Grid P4–5 / P6–10 / P11+	26.4% / 7.8% / 1.3%
Qualifying 1–3 / 4–6 / 7–10 / 11+	63.9% / 24.5% / 5.4% / 1.3%
Constructor leading the championship / 2nd / 3rd	58.9% / 39.9% / 26.2%
Driver form: 0% / >50% podiums in last 5	2.9% / 56.8%

The strongest single correlations with the target are driver and constructor podium form (0.55 each), qualifying position (−0.49) and grid (−0.48). Grid and qualifying position are strongly correlated with each other; both are retained and their added value is tested by ablation (Section 8). `grid_penalty` has almost no linear correlation (0.01) but is kept because its effect may be non-linear.

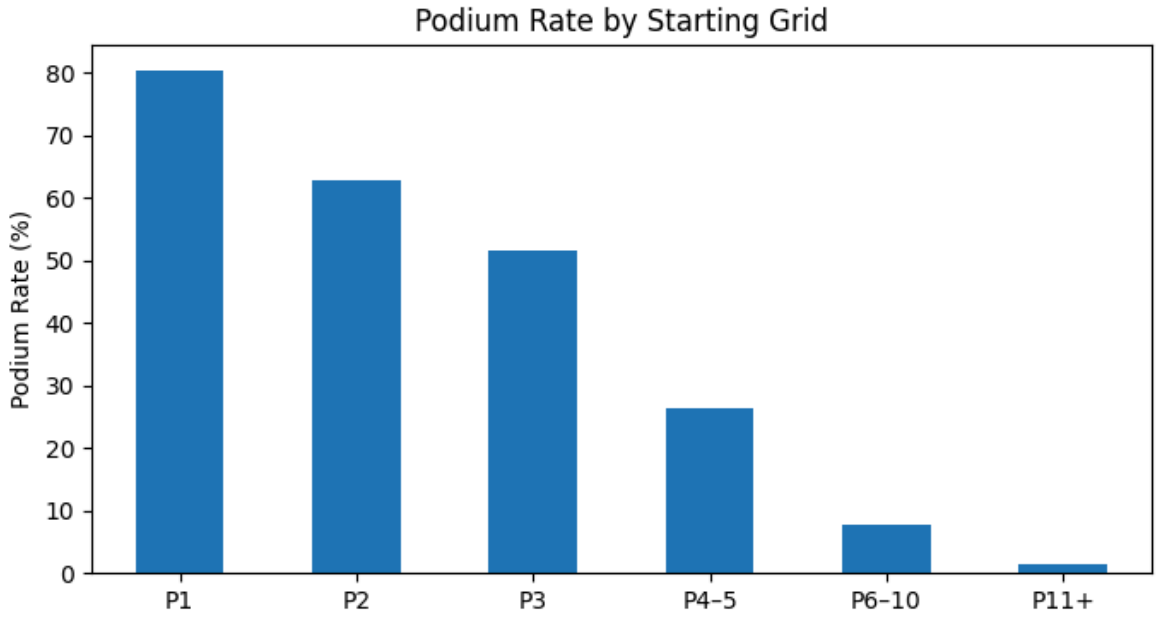


Figure 11. Podium rate by starting grid

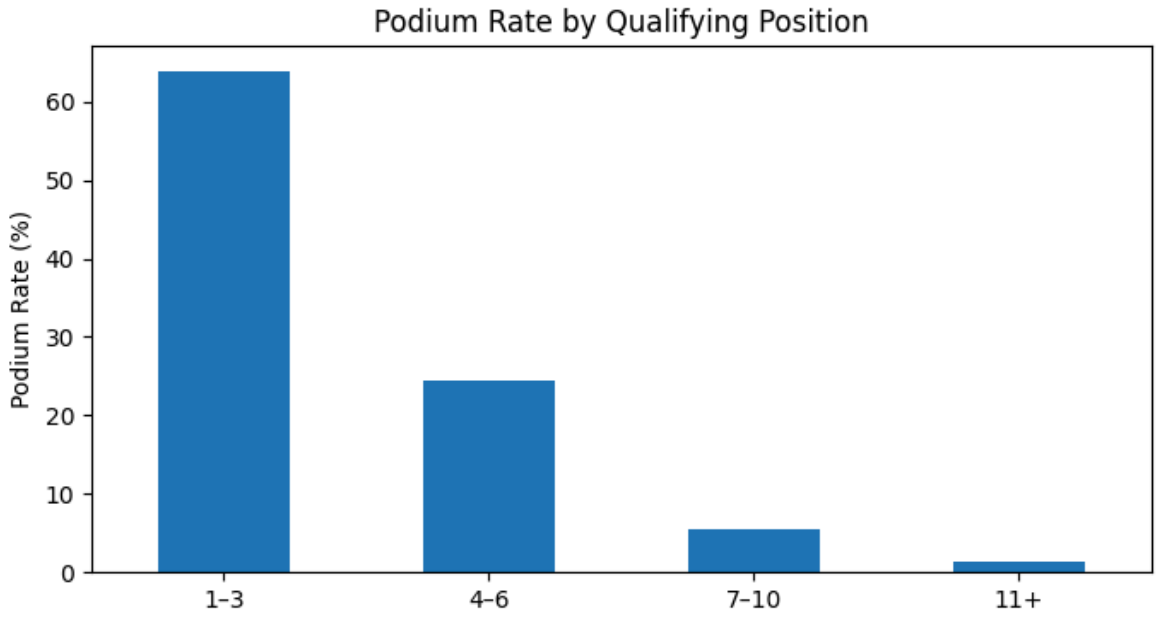


Figure 12. Podium rate by qualifying position

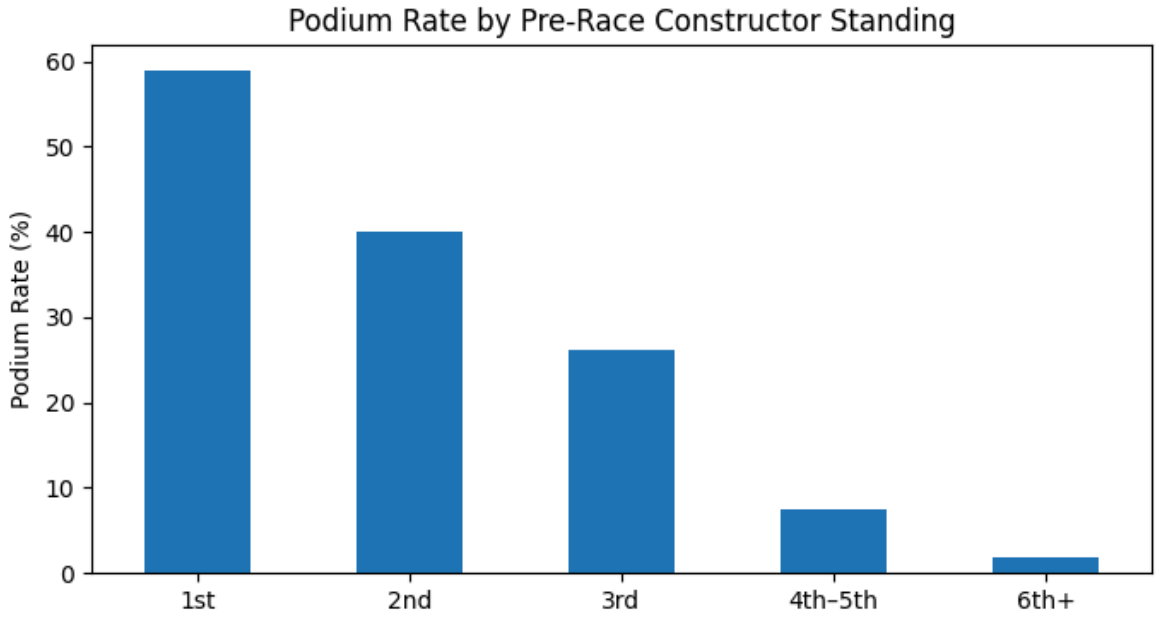


Figure 13. Podium rate by pre-race constructor standing

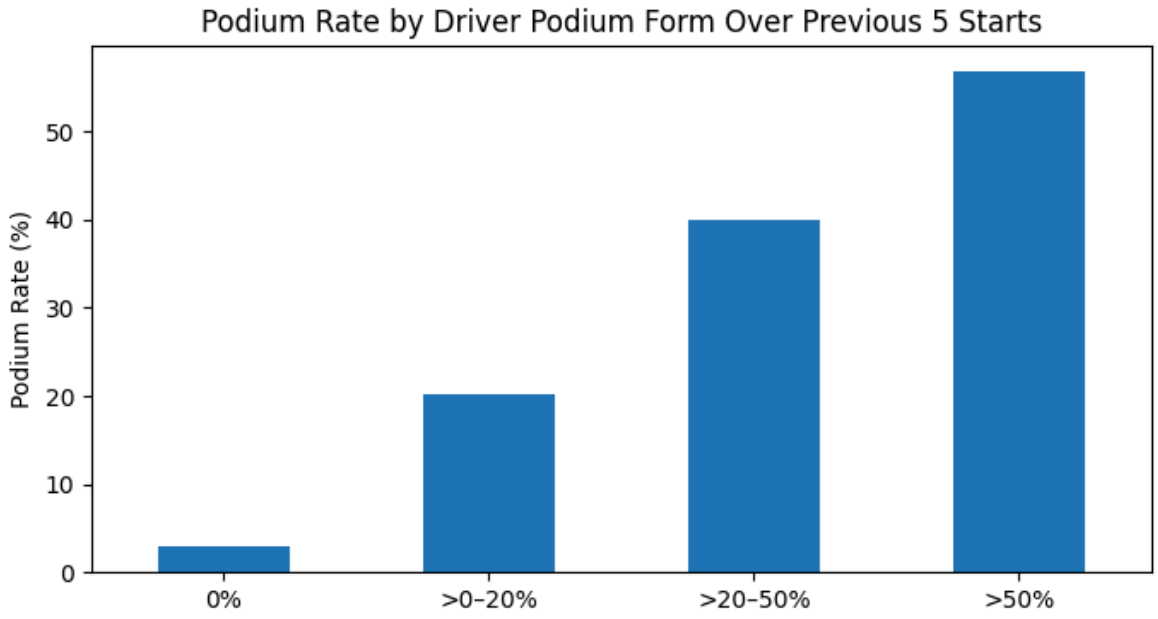


Figure 14. Podium rate by driver form over previous 5 starts

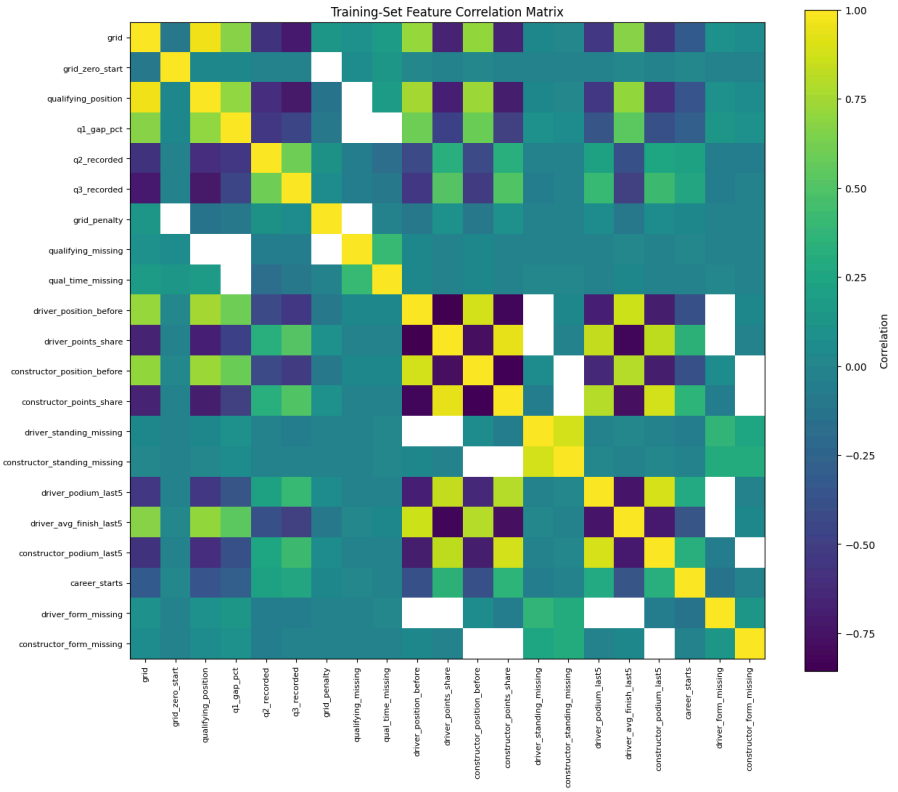


Figure 15. Training-set feature correlation matrix

6. Dataset split and preprocessing

The data is split by season, and all preprocessing is fitted on the training seasons only.

Set	Seasons	Rows	Podium rate
Train	2003–2019	6,800	14.2%
Validation	2020–2021	778	15.0%
Test	2022–2024	1,353	15.1%

Why a chronological split. The real task is to predict future races from past ones. A random split would mix rows from the same season into training and evaluation, letting the model learn that season's competitive order — information unavailable in advance. Whole seasons stay together, so no race is split across sets (asserted in the notebook).

Why training starts in 2003. Qualifying coverage is complete only from 2003 (Section 2). Earlier seasons would require inventing most qualifying values.

Caveats. The validation set is small and includes the shortened 2020 COVID season. The test period starts with the 2022 technical-regulation change, so it is a genuine era shift and an intentionally hard test.

Preprocessing

- **Q1 outlier cap** at the training 99th percentile (7.6%; raw training maximum 33.3%).
- **Median imputation** for the remaining missing values (train: standings about 6–7%, Q1 pace 1.8%, form about 1%). Missingness indicators are kept so the models can tell imputed values from observed ones.
- **Standardization** to zero mean and unit variance for Logistic Regression and the neural network. Gradient Boosting uses the imputed but unscaled values.

Both the imputer and the scaler are fitted on the training set and applied unchanged to validation and test. Standardization changes scale but not ordering: for example `career_starts` (0–321) and `grid` (0–24) end up on comparable scales.

Training feature	Range before	Mean (std) before	Range after	Mean (std) after
`grid`	0 – 24	11.13 (6.25)	−1.78 – 2.06	0 (1)
`qualifying_position`	1 – 24	11.16 (6.22)	−1.64 – 2.07	0 (1)
`q1_gap_pct` (after cap)	0 – 7.60	1.74 (1.53)	−1.15 – 3.88	0 (1)
`driver_points_share`	0 – 1	0.26 (0.30)	−0.83 – 2.56	0 (1)
`constructor_points_share`	0 – 1	0.30 (0.34)	−0.89 – 2.16	0 (1)
`career_starts`	0 – 321	92.9 (79.2)	−1.17 – 2.88	0 (1)

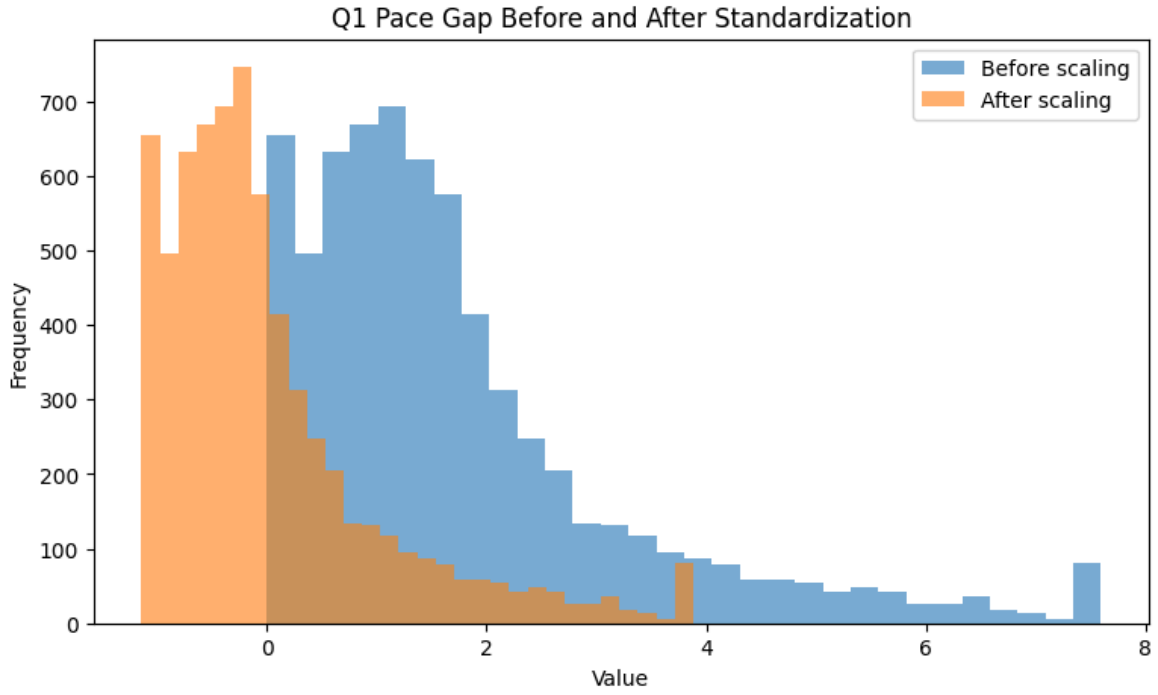


Figure 16. Q1 pace gap before and after standardization

Preprocessing-order experiments

Pipeline (Logistic Regression)	Validation ROC-AUC	Validation PR-AUC
Correct: imputer and scaler fitted on train only	0.9316	0.7315
Leaky: scaler fitted before the split	0.9314	0.7307
Leaky: imputer fitted before the split	0.9315	0.7310

The three variants are practically identical (PR-AUC differences below 0.001), so preprocessing leakage has little numerical effect on this dataset. The leaky versions remain methodologically invalid because validation information shapes the training transformation, so every reported model uses the train-only pipeline.

7. Models and evaluation

Three model families were trained on identical rows and the same 21 features. Logistic Regression and the neural network were statistically tied on validation; Gradient Boosting was strongest on the unseen test set.

Metrics

We report five metrics for every model: ROC-AUC, PR-AUC and F1 as required, plus precision and recall because F1 combines them. **PR-AUC is the primary metric** because podiums are a minority class (about 14–15% of rows) and PR-AUC focuses on how well true podium finishers are ranked above the rest. **ROC-AUC is secondary**, since it can look high even for weak models when positives are rare. F1, precision and recall use a probability threshold chosen to maximize F1 **on validation only**, then applied unchanged to test.

Models

- **Logistic Regression (baseline).** Models the log-odds of a podium as a weighted sum of the standardized features. Simple and interpretable, but assumes each feature acts linearly, so it cannot capture effects such as P1→P2 mattering more than P15→P16. Validation threshold 0.44.
- **Gradient Boosting.** Builds a sequence of shallow trees, each correcting the previous trees' errors, so it captures non-linear effects and interactions. More prone to overfitting and less interpretable. The default model overfit (train PR-AUC 0.80 vs validation 0.65); of four tuned configurations, GB4 (200 trees, learning rate 0.03, depth 2) had the best validation PR-AUC (0.708). Threshold 0.35.
- **Shallow neural network.** Feed-forward network with ReLU hidden layers and a sigmoid output, trained with Adam and early stopping on validation PR-AUC (patience 15, best weights restored). Early stopping is its only regularization. Of three architectures, one hidden layer of 32 units (NN1) was best; [32, 16] and [64, 32] did not help. Threshold 0.34.
- **Grid Baseline (reference).** Ranks drivers by actual starting grid for ROC-AUC and PR-AUC (grid 0 ranked below every normal position) and predicts a podium for P1–P3 starters for F1, precision and recall; no learning.

Seed robustness. Across five random seeds, NN1 reached validation PR-AUC 0.7334 ± 0.0057, versus 0.7315 for Logistic Regression. The difference (0.002) is smaller than the seed-to-seed spread, so the two are **effectively tied**. The seed-42 run was fixed as the representative network before testing.

Complete results

Model	Split	ROC-AUC	PR-AUC	F1	Precision	Recall
Grid Baseline	Train	0.909	0.628	0.647	0.649	0.646
Grid Baseline	Validation	0.909	0.639	0.692	0.692	0.692
Grid Baseline	Test	0.914	0.647	0.649	0.650	0.647
Logistic Regression	Train	0.930	0.689	0.652	0.659	0.646
Logistic Regression	Validation	0.932	0.732	0.695	0.698	0.692
Logistic Regression	Test	0.936	0.709	0.644	0.685	0.608
Gradient Boosting	Train	0.940	0.729	0.694	0.631	0.771
Gradient Boosting	Validation	0.928	0.708	0.697	0.669	0.727
Gradient Boosting	Test	**0.940**	**0.724**	**0.697**	0.647	**0.755**
Shallow NN (NN1)	Train	0.939	0.708	0.680	0.632	0.736
Shallow NN (NN1)	Validation	0.930	0.735	0.705	0.686	0.727
Shallow NN (NN1)	Test	0.936	0.694	0.668	0.643	0.696

Grid Baseline. For ROC-AUC and PR-AUC, actual grid position is used as an ordered ranking score, with better starting positions receiving higher scores. For F1, precision and recall, the simple classification rule predicts a podium for drivers starting P1–P3. This provides both a ranking and hard-classification reference for the learned models.

Interpretation

- **No overfitting** for Logistic Regression or NN1 (train and validation scores are close); tuning removed Gradient Boosting's overfitting.
- **On test**, Gradient Boosting was strongest (PR-AUC 0.724, F1 0.697) and its PR-AUC even rose from validation. NN1 dropped most (−0.042 PR-AUC), showing its small validation edge did not transfer to the new regulation era.
- **Against the Grid Baseline:** the Grid Baseline achieved test PR-AUC = 0.6474 and F1 = 0.6486. Logistic Regression improved ranking performance to PR-AUC = 0.7088, although its fixed validation-selected threshold produced slightly lower test F1 = 0.6442. Gradient Boosting provided the strongest improvement over the grid-only baseline, reaching test PR-AUC = 0.7243 and F1 = 0.6968. This shows that starting grid alone is already a strong predictor, while the engineered qualifying, standings and form features provide additional predictive ranking information.
- **Complexity did not clearly pay off:** the simple Logistic Regression baseline was tied with the network on validation and close to Gradient Boosting on test.

No model, threshold or feature was changed after looking at test results.

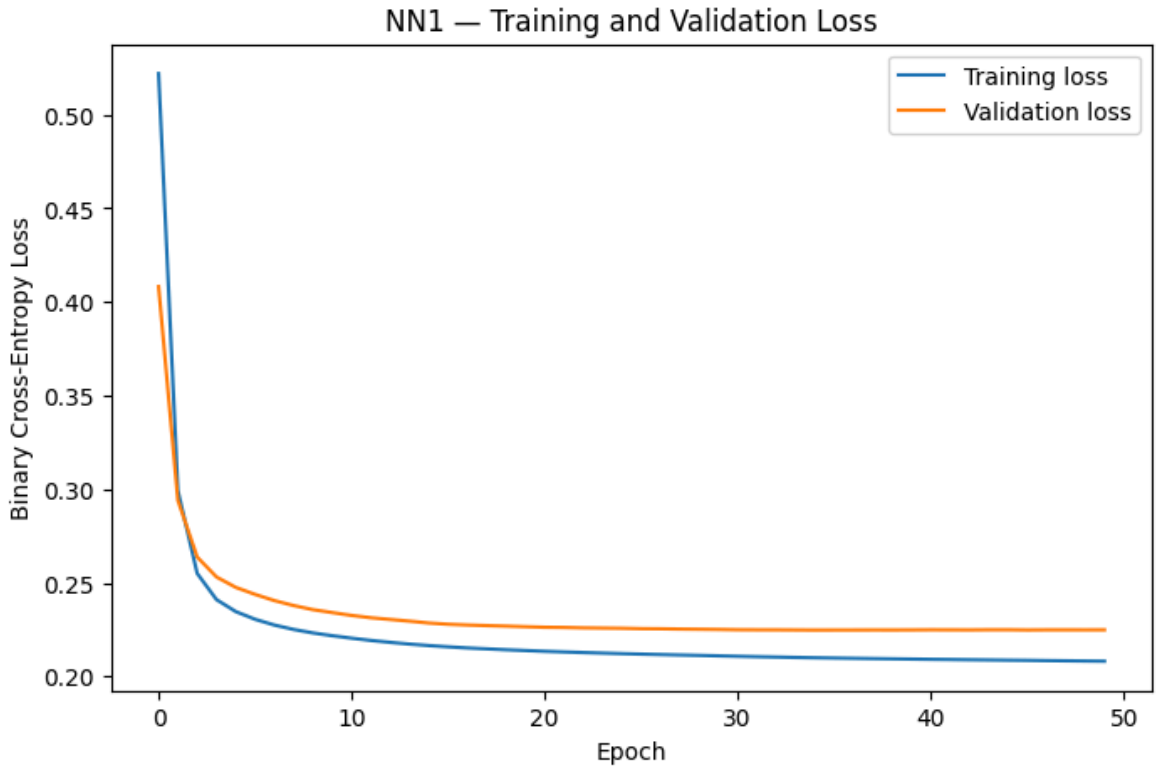


Figure 17. Neural network training and validation loss

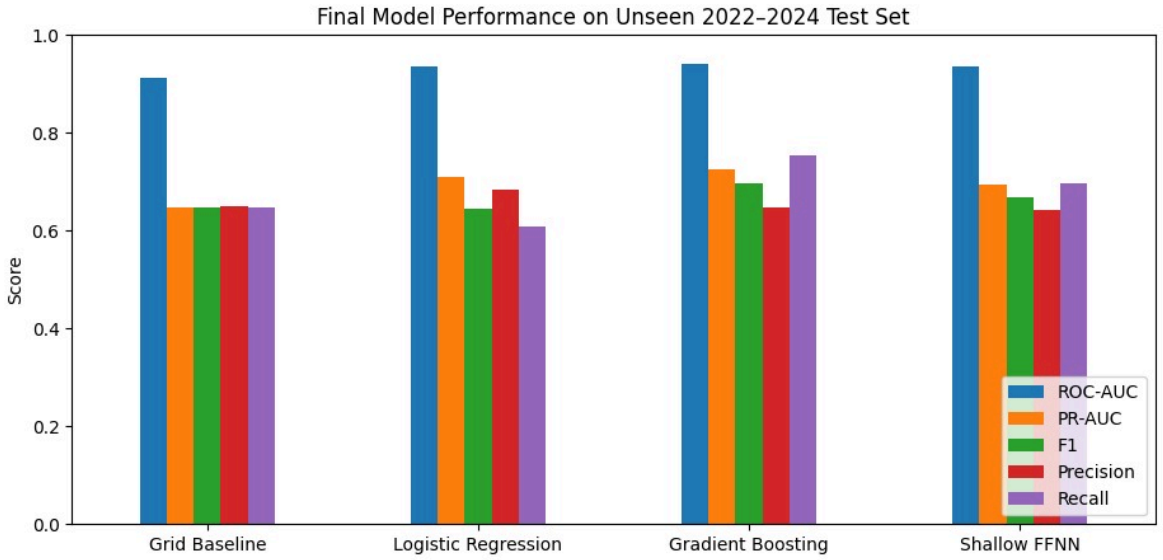


Figure 18. Final model performance on the unseen 2022–2024 test set

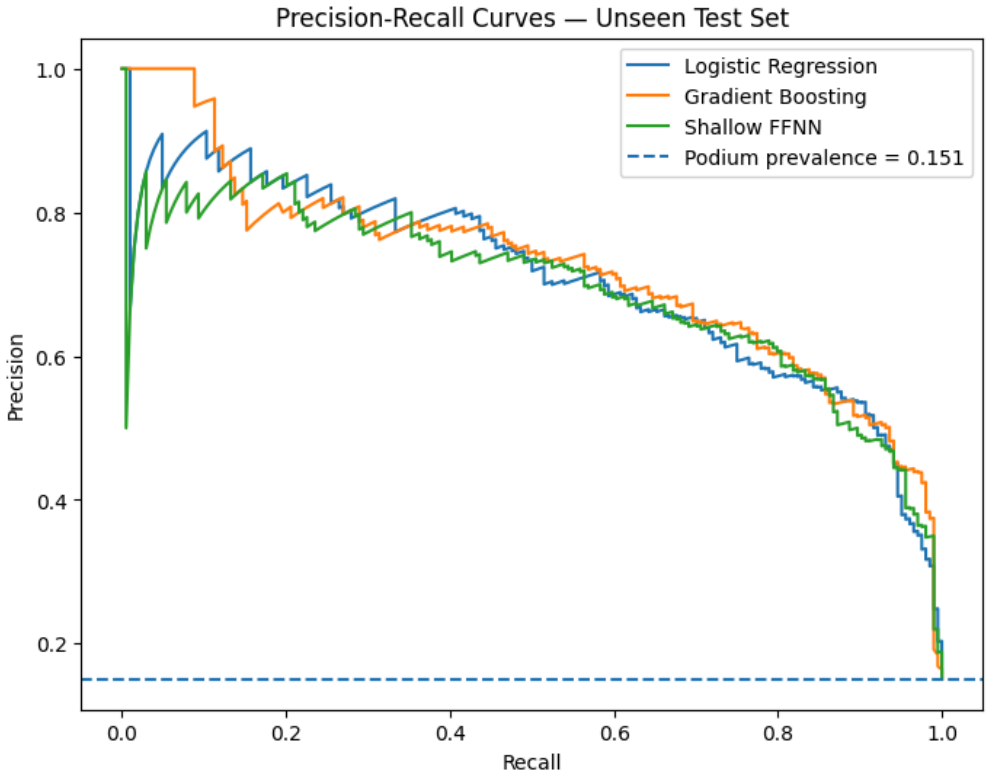


Figure 19. Precision-recall curves on the unseen test set

Validation diagnostics per model

Threshold curves show how F1, precision and recall change with the cutoff on validation; the dashed line marks the selected threshold. Confusion matrices and precision-recall curves are at that threshold.

Logistic Regression

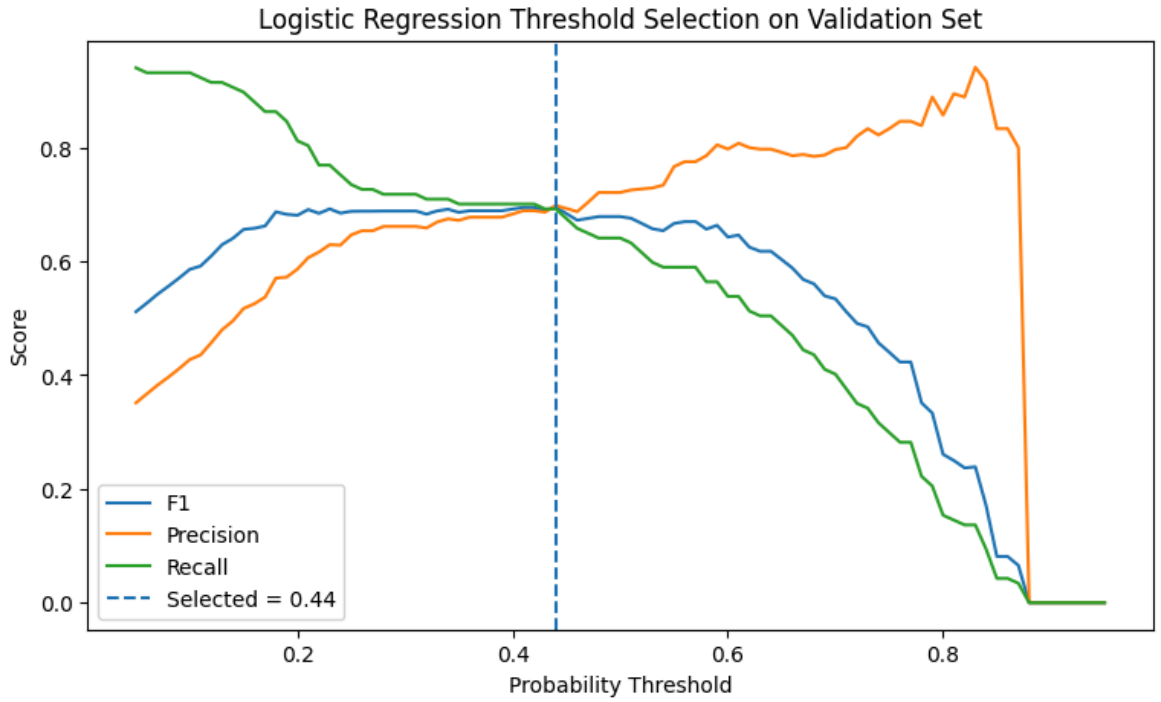


Figure 20. Logistic Regression threshold selection on validation

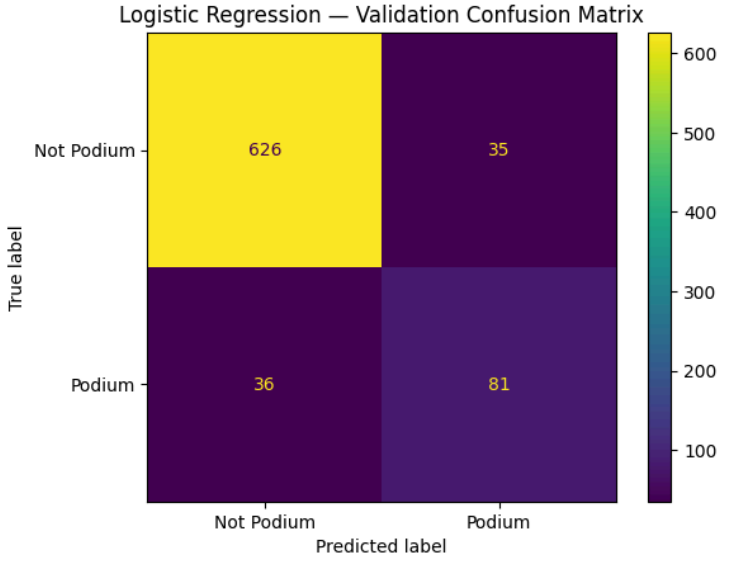


Figure 21. Logistic Regression validation confusion matrix

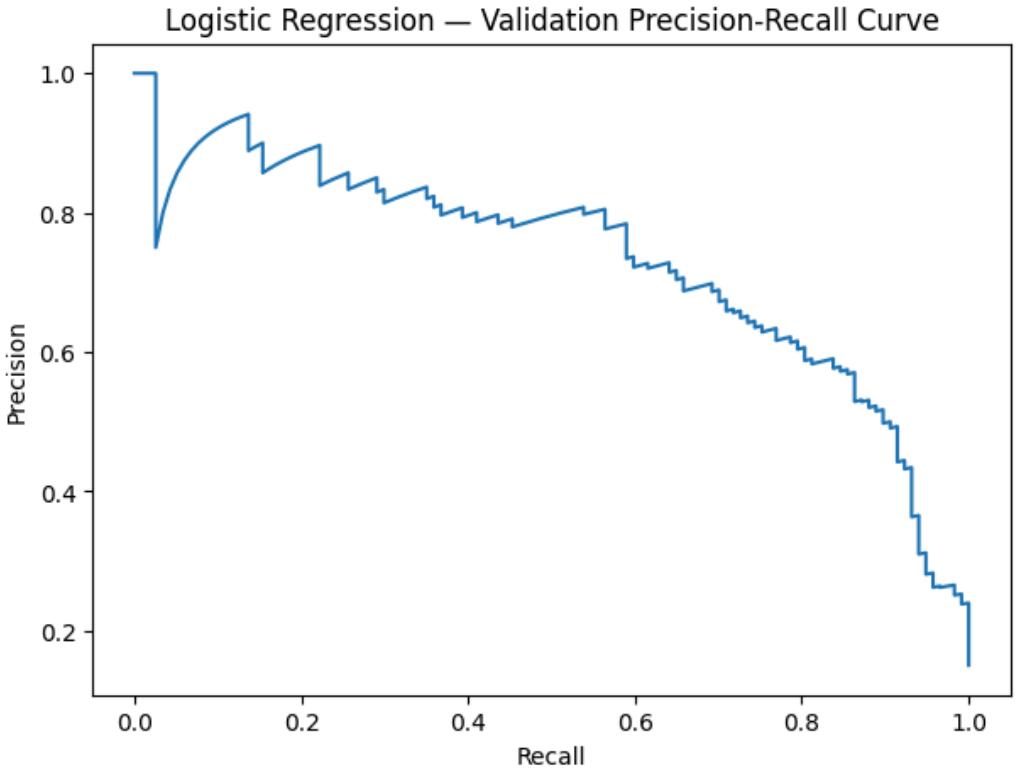


Figure 22. Logistic Regression validation precision-recall curve

Gradient Boosting (GB4)

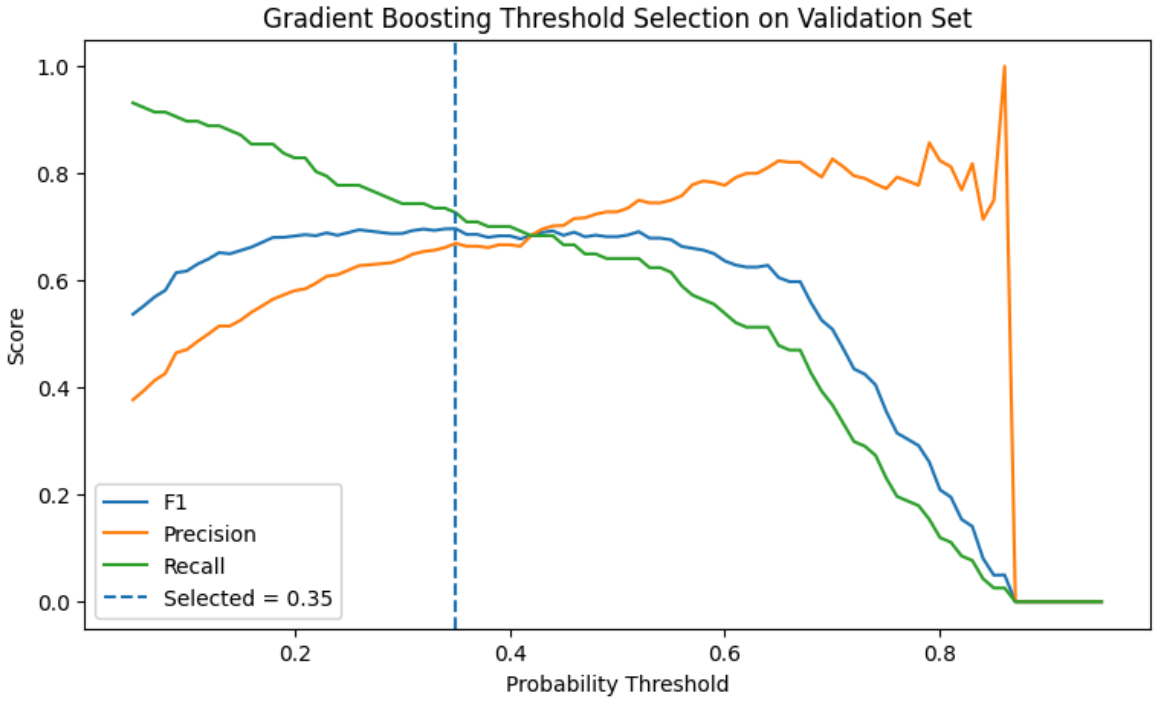


Figure 23. Gradient Boosting threshold selection on validation

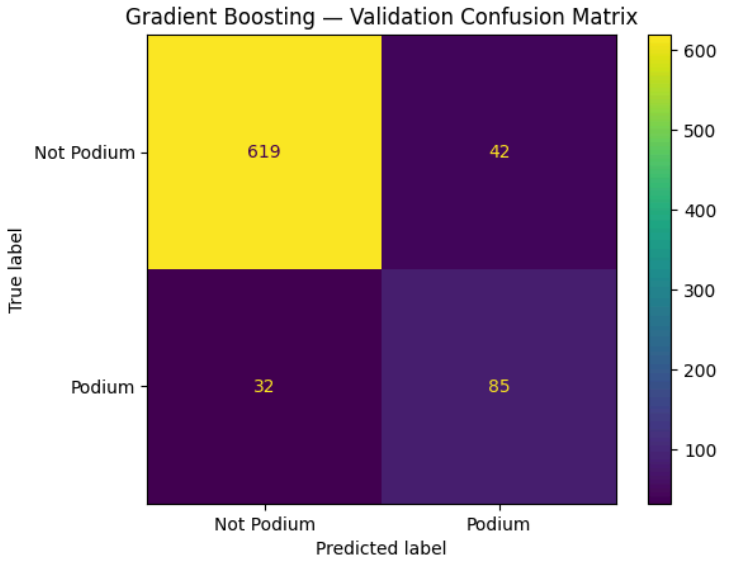


Figure 24. Gradient Boosting validation confusion matrix

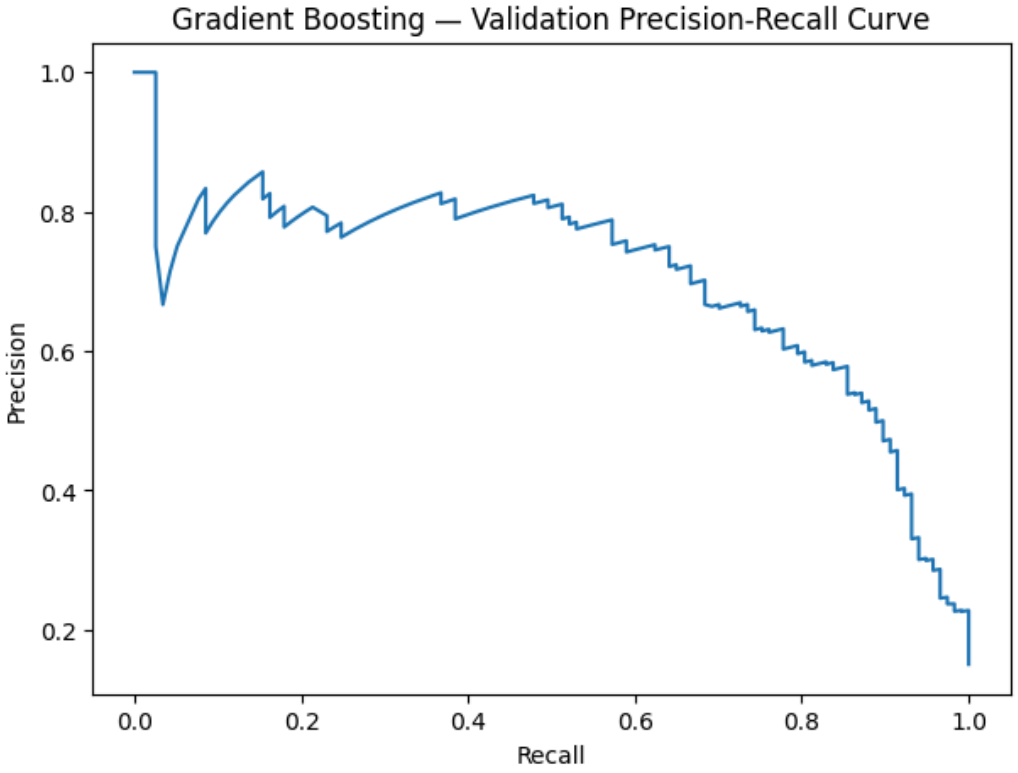


Figure 25. Gradient Boosting validation precision-recall curve

Shallow neural network (NN1)

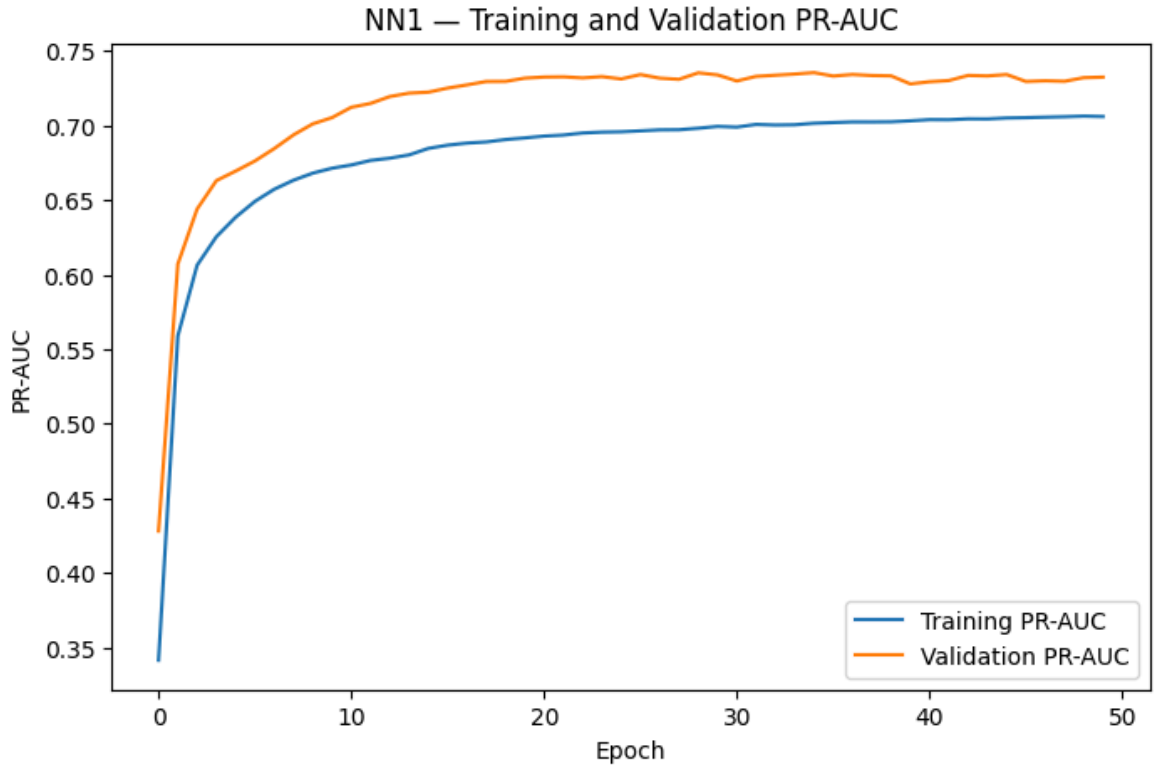


Figure 26. NN1 training and validation PR-AUC by epoch

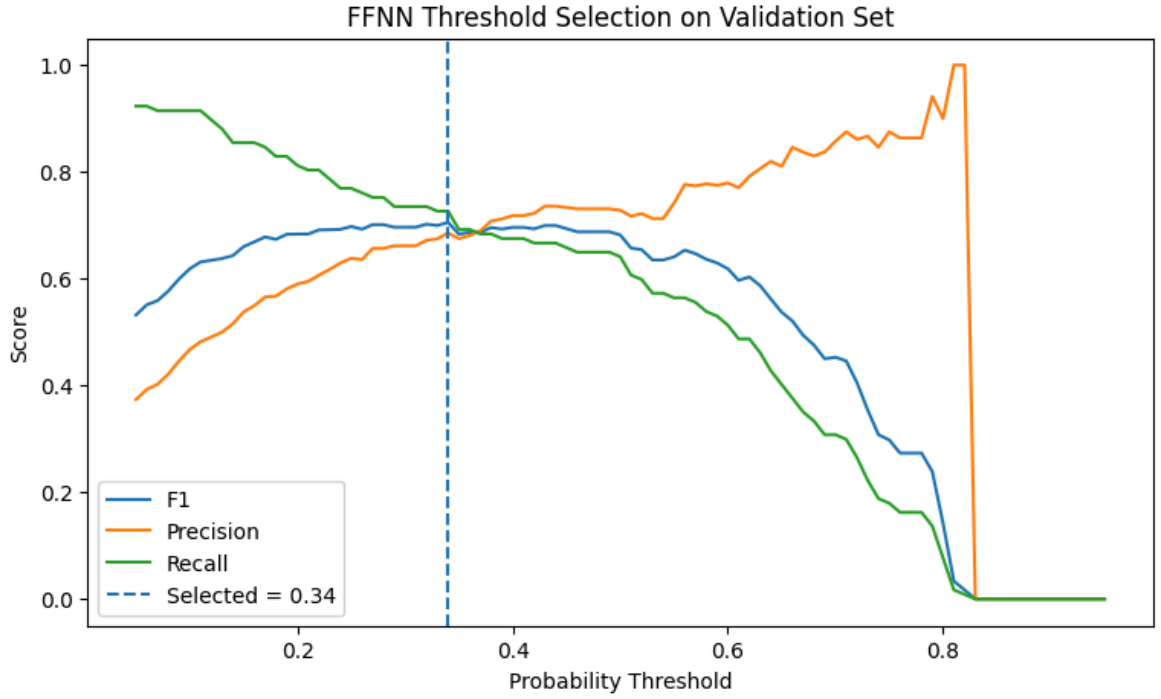


Figure 27. NN1 threshold selection on validation

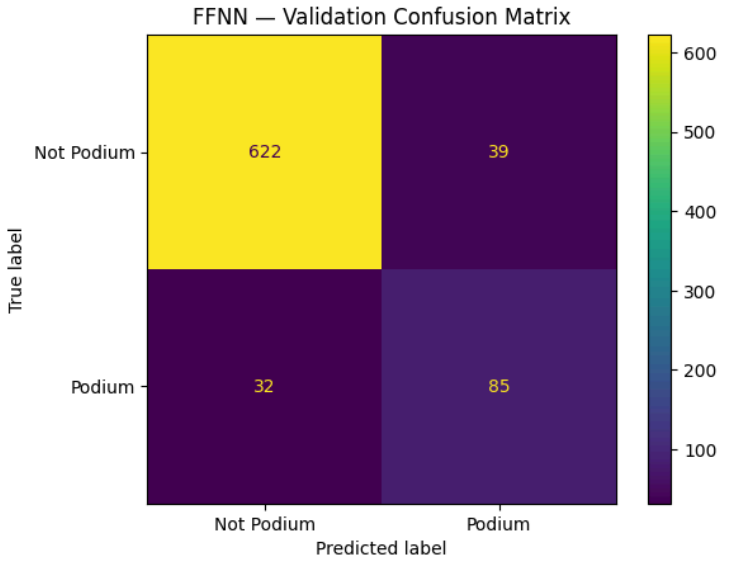


Figure 28. NN1 validation confusion matrix

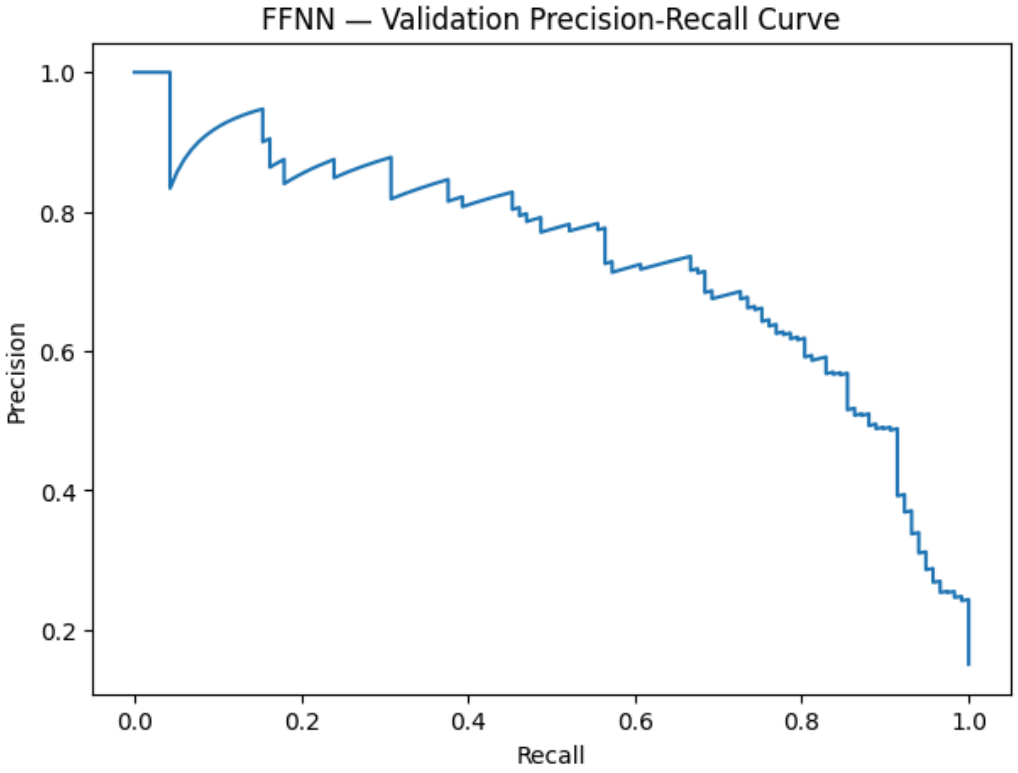


Figure 29. NN1 validation precision-recall curve

Validation comparison

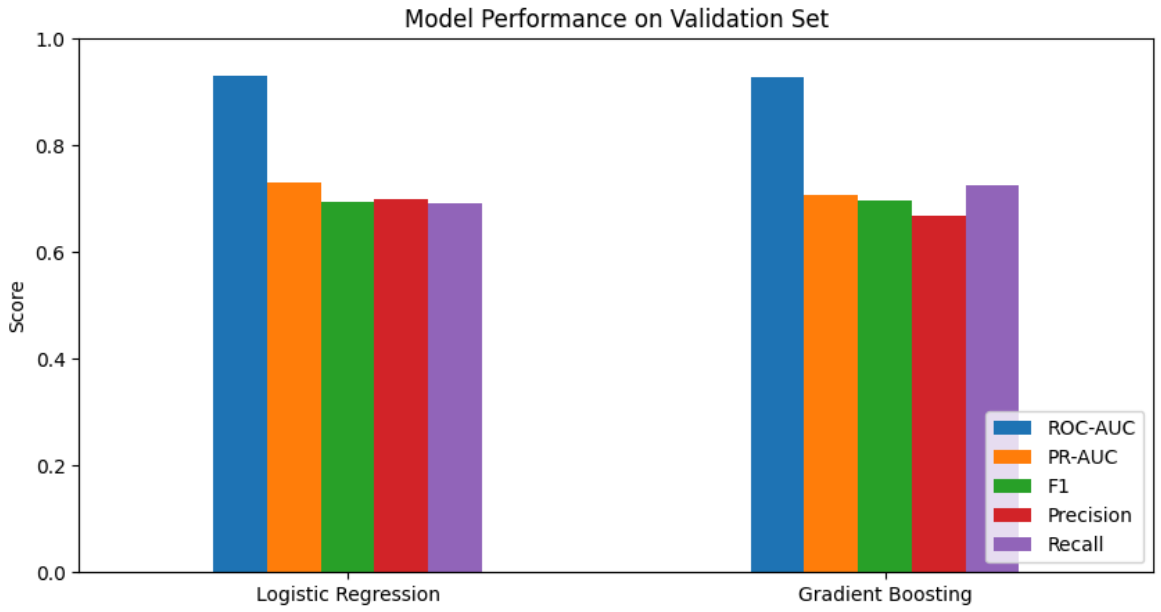


Figure 30. Validation performance, Logistic Regression vs Gradient Boosting

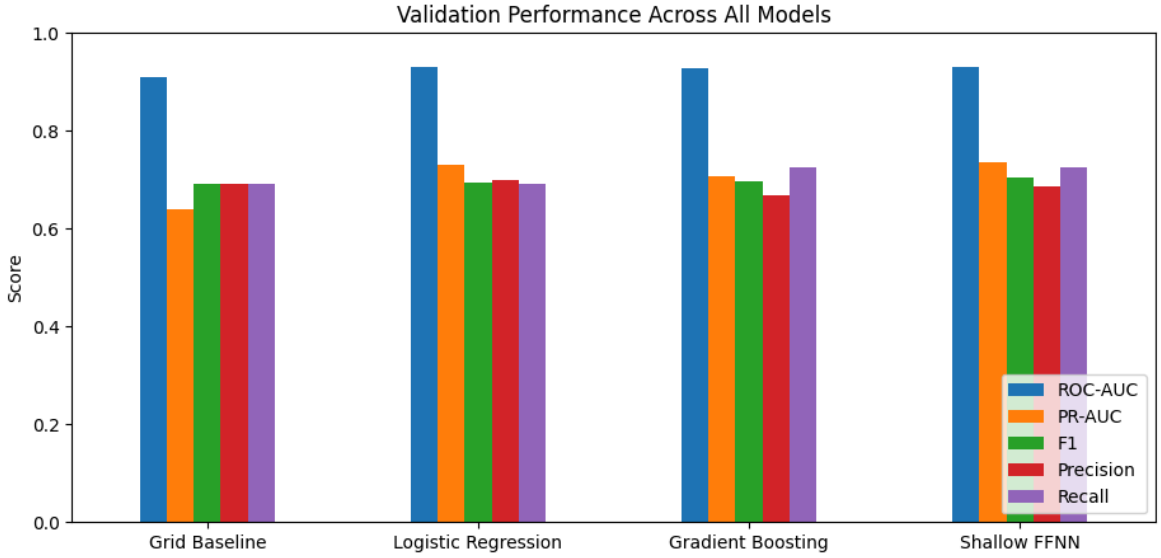


Figure 31. Validation performance across all models

Test diagnostics

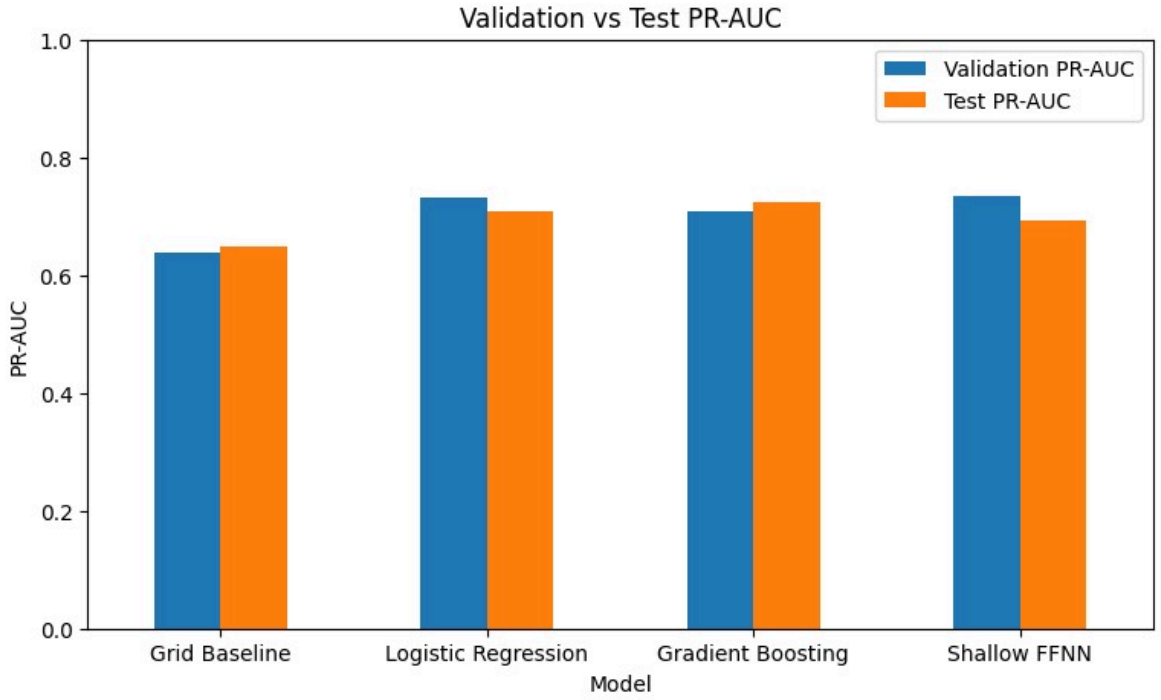


Figure 32. Validation vs test PR-AUC



Figure 33. Logistic Regression test confusion matrix

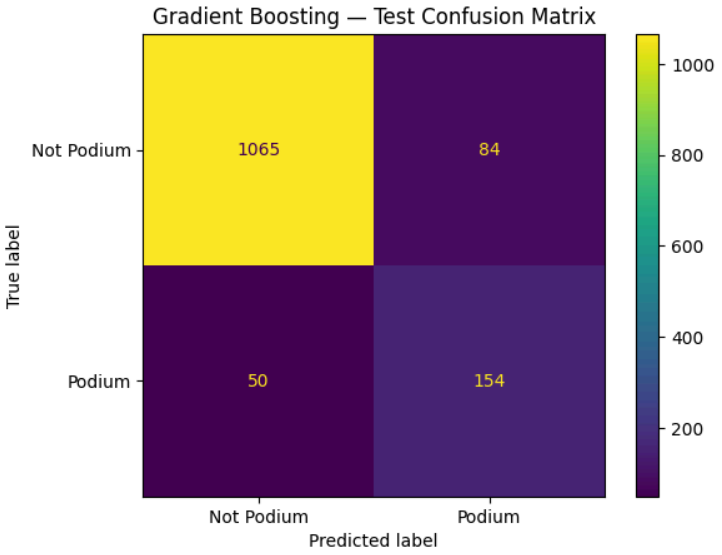


Figure 34. Gradient Boosting test confusion matrix

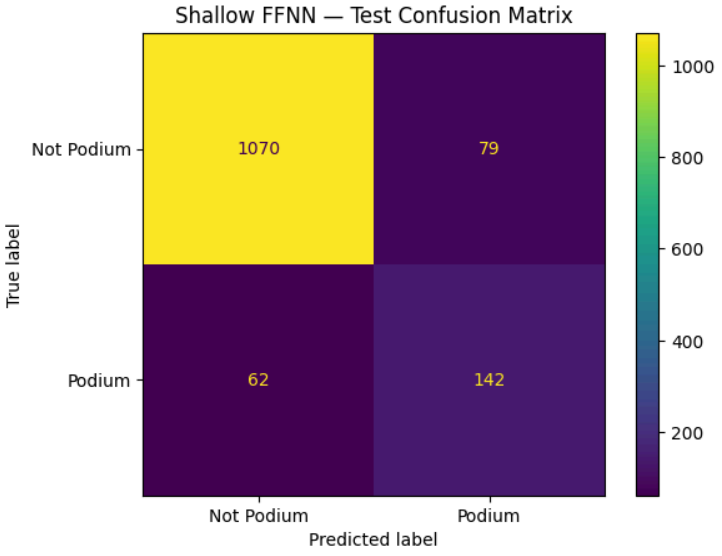


Figure 35. NN1 test confusion matrix

8. Feature-group ablation

Removing any single group lowers validation PR-AUC, so all four groups are retained; grid and qualifying matter most for the network, qualifying and standings for Logistic Regression.

Each group was removed in turn and the model retrained, with imputation and scaling refitted on the remaining training features. The NN1 architecture and settings were fixed, and each NN experiment was repeated over five seeds.

Removed group	NN1 mean validation PR-AUC (± std)	NN1 change vs all features	Logistic Regression PR-AUC	Logistic Regression change
None (all 21 features)	0.7334 ± 0.0057	—	0.7315	—
Grid	0.7172 ± 0.0137	−0.0162	0.7296	−0.0019
Qualifying	0.7176 ± 0.0080	−0.0158	0.7078	−0.0237
Standings	0.7225 ± 0.0115	−0.0110	0.7203	−0.0112

Removed group	NN1 mean validation PR-AUC (± std)	NN1 change vs all features	Logistic Regression PR-AUC	Logistic Regression change
Form	0.7290 ± 0.0034	−0.0044	0.7276	−0.0039

Findings.

- For the network, grid and qualifying produce the largest average drops, then standings; form adds the least.
- The seed-to-seed spread is comparable to these drops, so the NN ablation shows that each group contributes, not a precise ranking.
- The deterministic Logistic Regression ablation relies most on qualifying, then standings. Removing grid barely hurts it because grid and qualifying position are strongly correlated and a linear model can substitute one for the other.
- Feature importance is therefore model-dependent. Since every group helps at least one model and all are valid pre-race information, all four are kept.

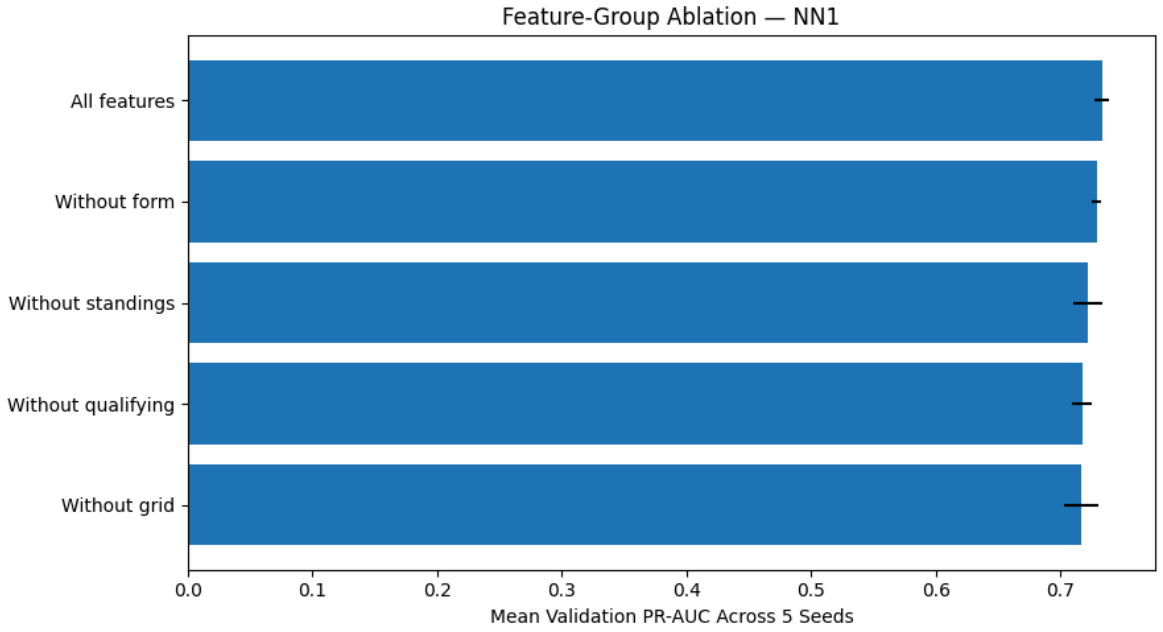


Figure 36. Feature-group ablation for NN1, mean of 5 seeds

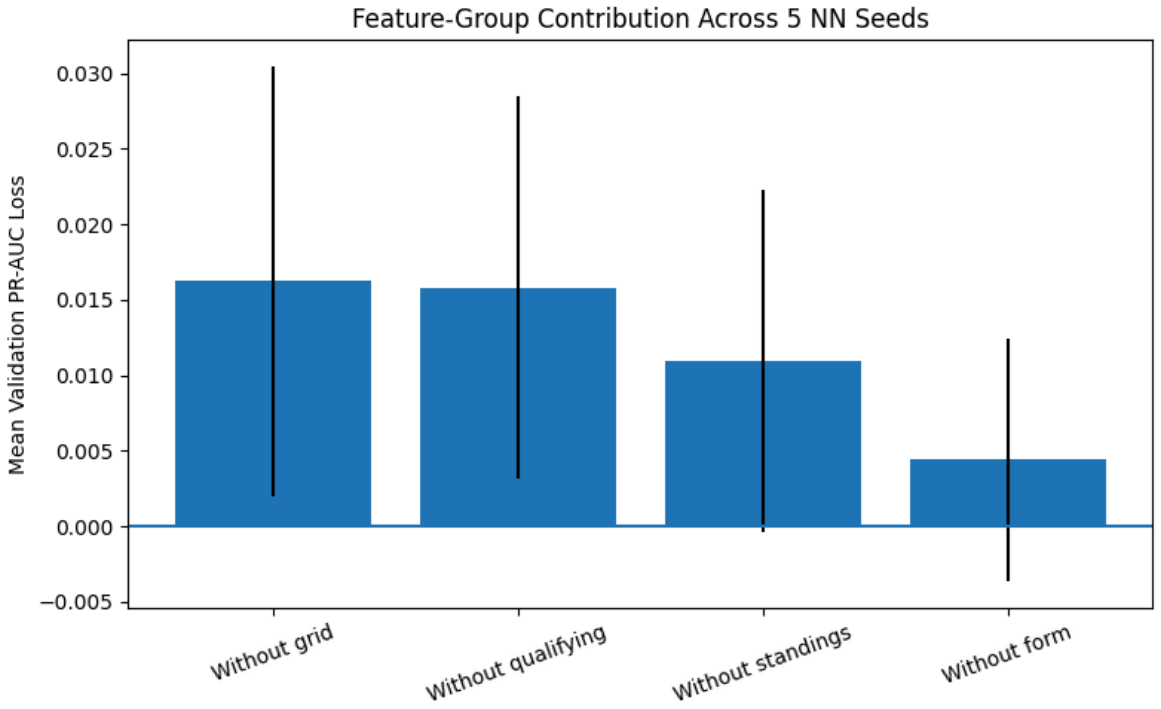


Figure 37. Feature-group contribution across 5 NN seeds

9. Explainability (XAI)

Permutation importance and SHAP agree that the network relies mainly on grid and qualifying position, followed by recent driver form and championship strength.

Explanations are not causal. A feature being important to the model only means the model relies on it to make predictions. It is not evidence that the feature causes podium finishes. Every finding below describes associations the model learned.

All explanations are computed for NN1, the representative network carried forward from validation, on validation data.

Global: permutation importance

Each feature was shuffled 10 times and the average drop in validation PR-AUC (baseline 0.7354) recorded.

Feature	Mean PR-AUC drop when shuffled
`grid`	0.149
`qualifying_position`	0.148
`driver_podium_last5`	0.071
`driver_points_share`	0.068
`driver_standing_missing`	0.026
`grid_zero_start`	0.025
`constructor_points_share`	0.024

All other features cause drops below 0.02.

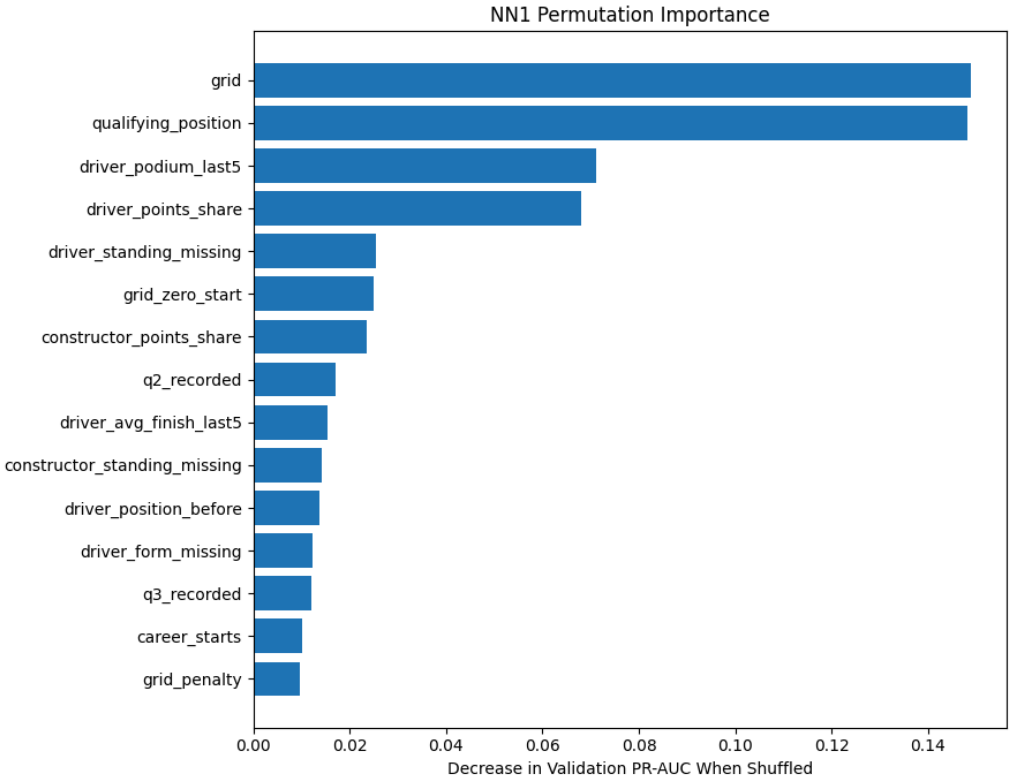


Figure 38. NN1 permutation importance

Global: SHAP

SHAP (permutation explainer, 50-row training background, 100 validation rows explained) gives a similar ranking: qualifying position and grid have the largest mean absolute SHAP values, followed by driver podium form, constructor points share, driver points share and driver championship position. The beeswarm shows sensible directions: better grid and qualifying positions, stronger recent podium form and higher points shares all push predictions toward a podium.

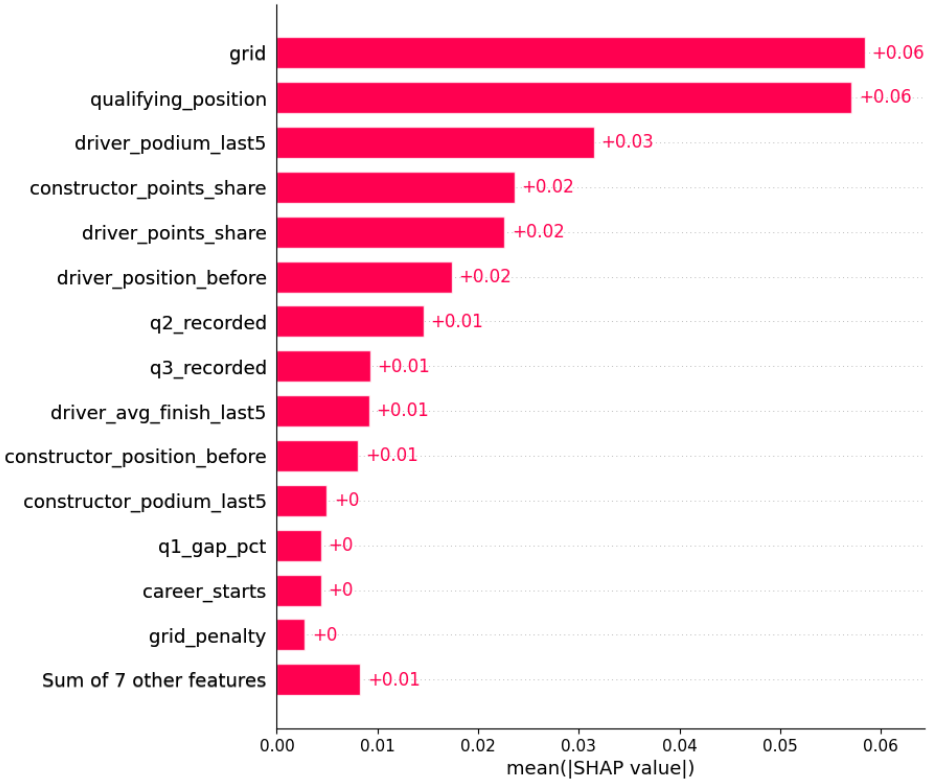


Figure 39. SHAP mean absolute values, top 15 features

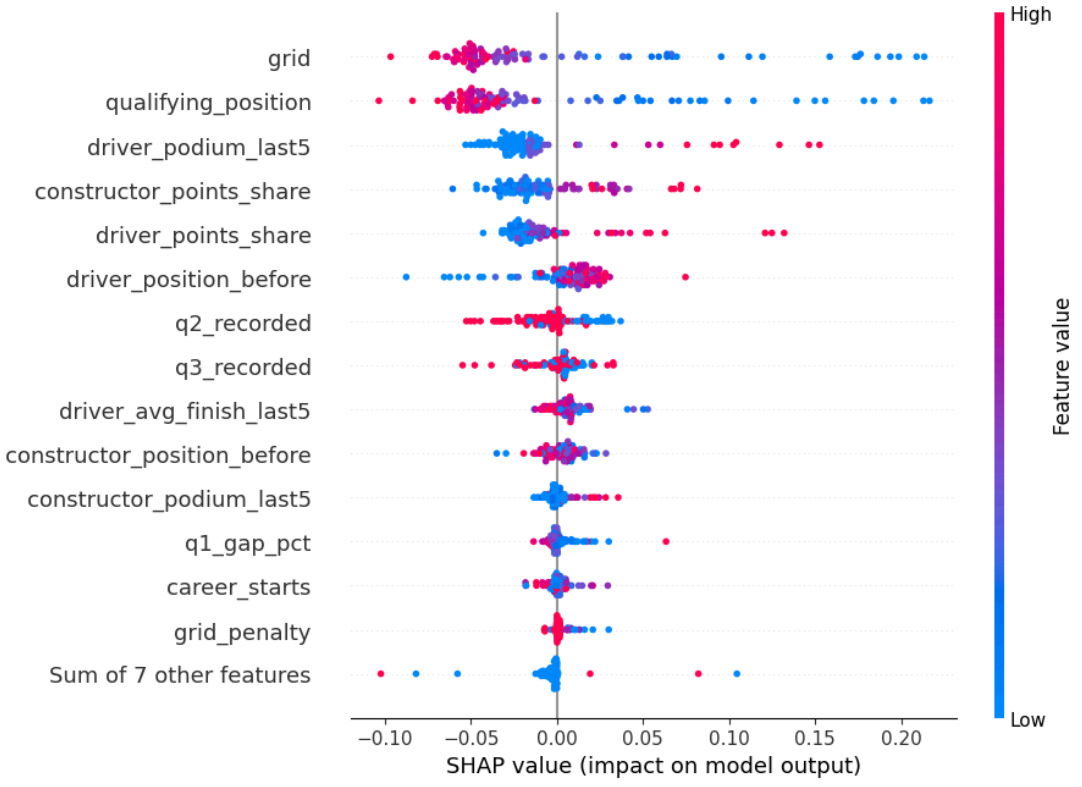


Figure 40. SHAP beeswarm, top 15 features

Local: LIME

LIME explained one borderline but correct prediction: a 2021 driver (round 9) who started and qualified 5th, had podiums in 40% of their previous five starts, and drove for the second-placed constructor holding 84% of the leader's points. NN1 gave a podium probability of 0.341, just above the 0.34 threshold, and the driver did finish on the podium.

LIME condition	Weight toward podium
`grid` ≤ 6	+0.112
`qualifying_position` ≤ 6	+0.103
`grid_zero_start` = 0	+0.094
`driver_podium_last5` > 0.20	+0.082
`constructor_points_share` > 0.47	+0.065
`qual_time_missing` = 0	−0.169
`constructor_standing_missing` = 0	−0.113

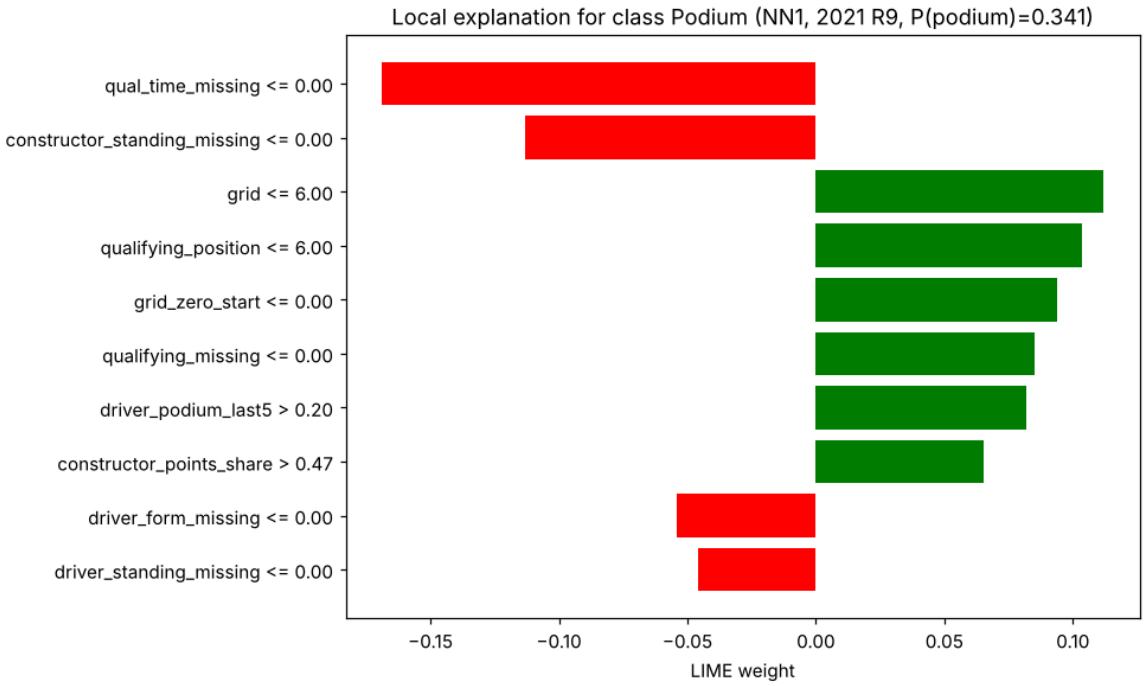


Figure 41. LIME local explanation for the 2021 round 9 podium prediction

The positive drivers (front-half grid and qualifying, good recent form, a strong constructor) match the global picture. The negative weights on missingness indicators, for data that was actually present, should be read cautiously: LIME fits a local surrogate model, and correlated or binary features can receive unintuitive weights.

10. Inference function

`predict_podium(record)` takes one driver's pre-race feature values and returns NN1's podium probability and class, using only objects fitted during training.

1. Arrange the input into the 21-feature layout (absent features become missing).
2. Rebuild the derived features exactly as in feature engineering: `grid_zero_start`, `grid_penalty` and all missingness indicators.
3. Apply the training Q1 pace cap (7.6%).
4. Apply the training-fitted median imputer, then the training-fitted scaler.
5. Predict with NN1 and apply the validation-selected threshold (0.34).

Nothing is refitted at inference time, and post-race variables are never accepted as inputs.

Demonstration	Missing inputs	Podium probability	Prediction	Actual
Complete record (2022 round 2, started 4th)	0	0.513	Podium	Podium
Same record with 5 features removed	5	0.912	Podium	Podium

The partial record removed Q1 pace, driver championship position, driver points share and both driver form features. Its probability rose, which does not mean missing data improves podium chances: removal replaces actual values (this driver had 0 points after a round-1 retirement) with training medians and switches on missingness indicators. Predictions on incomplete records should therefore be treated with more caution.

11. Known limitations

- **Qualifying-coverage drift.** Qualifying data is absent before 1994 and patchy until 2002, so the model only learns from 2003 onward and cannot be applied to earlier eras.
- **Era sensitivity.** Regulation changes (2010 points system, 2014 hybrid engines, 2022 ground-effect cars) shift what predicts a podium. The test set spans a new era, and NN1's PR-AUC fell 0.042 from validation to test.
- **Small validation set.** 778 rows, including the shortened 2020 season, make small metric differences unreliable; the NN and Logistic Regression could not be separated.
- **Fixed thresholds.** Thresholds tuned on 2020–2021 lost recall in 2022–2024, lowering F1 even where ranking quality held up.
- **Missing information.** No weather, tyre choice, practice pace, strategy or in-race incidents are available, so luck and reliability failures during the race cannot be predicted.
- **Constructor identity.** Renamed teams (Toro Rosso → AlphaTauri → RB, Force India → Racing Point → Aston Martin) restart their form and standings history.
- **Judgment calls.** The mechanical-retirement classification, nationality-to-country mapping and the outcome-based shared-drive rule are documented decisions; the last affects only pre-1965 rows outside the modeling period.
- **Association, not causation.** Q2's home-race result and all XAI findings describe associations only.

12. Conclusion

A validated, leakage-safe driver-race table built from 14 relational tables supports strong podium prediction from pre-race information alone. On the unseen 2022–2024 seasons all models reach ROC-AUC about 0.94 and PR-AUC 0.69–0.72. All learned models improve ranking quality over the grid-only baseline (test PR-AUC 0.647), with Gradient Boosting performing best on the unseen test period. Starting grid and qualifying carry most of the signal, with recent form and championship strength adding further value. Added model complexity brought little over the Logistic Regression baseline, and performance under regulation changes remains the main open risk.